ENGLISH FOR WORK / DESIGN / EVIDENCE / QUALITY / TRAINING

Medical Devices English

Specify the need. Explain the evidence. Keep the boundary clear.



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Eight practical lessons. Eleven extended conversations.
Vocabulary, phrase banks, guided speaking, and explained answers.

Revised 2026 edition / Upper-intermediate to advanced

Your route through the book

01 User Needs and Design Inputs, page 4

Turn an ambiguous design discussion into a scoped request for users, conditions, and measurable criteria.

02 Risk Management and Hazard Analysis, page 14

Separate a hazard, an exposure situation, possible harm, and the evidence for a risk-control claim.

03 Verification, Validation, and Design Review, page 26

Report completed specification checks without implying that intended-use evaluation or release authorization is complete.

04 Usability and Human Factors, page 37

Report observed use difficulties neutrally and distinguish exploratory findings from validation or population-level rates.

05 Regulatory Pathways and Submissions, page 49

Distinguish a proposed US submission pathway, supporting evidence, and an actual regulatory decision.

06 Complaints, MDRs, and Postmarket Signals, page 61

Capture a complaint accurately, route missing information promptly, and distinguish counts from rates and reports from causation.

07 Manufacturing, Suppliers, and Nonconformance, page 76

Separate affected-lot scope, observed failures, containment, cause, and authorized disposition in a supplier discussion.

08 Clinical Training and Labeling Boundaries, page 87

Acknowledge an out-of-scope product-use request, keep the demonstration within authorized materials, and arrange a precise follow-up.

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Suggested study range: B2-C1. This is an editorial study range, not a certified proficiency assessment.

A precise sentence protects the decision.

Device teams work across engineering, clinical, commercial, and regulatory perspectives. The same word can mean different things in different meetings: tested is not the same as validated, a complaint is not a confirmed cause, and a proposed pathway is not an authorization. These cases give you the language to make those distinctions without losing the practical next step.

Put the user beside the requirement

Name who will use the device, for which task, and in what environment. A broad adjective such as intuitive cannot replace a reviewable requirement or evidence from the relevant users.
"Which users and conditions does this requirement cover?"

Say what the evidence establishes

Identify the object, method, criteria, and result of a test. A passed specification check and evidence about intended use answer related but different questions.
"The output met this specification; the intended-user evaluation remains open."

Separate an observation from its explanation

Describe the event before naming a cause. Missing information is a follow-up task, not permission to replace uncertainty with a reassuring conclusion.
"The caller reported an interruption; the cause has not been established."

State whose decision it is

Routing a question, proposing a change, and authorizing release are different acts. Use a named review route and a visible status instead of implying that discussion has settled an approval.
"We can propose the change; the designated reviewer must assess it."

Professional boundaries

Original fictional language practice, not device operating instructions or medical, legal, quality-system, or regulatory advice. US regulatory examples are identified as such; other jurisdictions differ. Use the current labeling, applicable requirements, trained personnel, and your organization's controlled procedures. No scenario authorizes patient care or product release.

A Read the situation

LESSON 01 / USER NEEDS AND DESIGN INPUTS

Turn an ambiguous design discussion into a scoped request for users, conditions, and measurable criteria.

> **What does easy to use mean?**
>
> Design engineer Mira and product lead Owen review a fictional monitoring-device brief. It says easy to use and suitable everywhere. The agreed initial audience is trained clinic staff, but an appended slide also mentions home caregivers. No home-use assessment is supplied. A proposed setup-time target of two minutes has not been reviewed or approved. Mira and Owen must distinguish the agreed scope, the proposed expansion, and the missing basis for a measurable design input.

People: Mira, Design engineer; Owen, Product lead.

Read the record

Choose one answer per question using the facts above.

1. Who is the agreed initial audience?

A. Children using the device independently
B. Trained clinic staff
C. All home caregivers
D. Every possible user without training

2. What is the status of the two-minute target?

A. Proposed, not reviewed or approved
B. Approved and already verified
C. Required by a supplied regulation
D. Demonstrated in a home-use study

3. What additional evidence question follows from adding home caregivers to the brief?

A. Whether the existing clinic-user results have already established identical home performance
B. Whether mentioning home use automatically adds it to the approved baseline
C. Whether a numeric setup target alone removes the need to assess the new users
D. Whether the new users, training, tasks, and home conditions support the proposed use

Answers and explanations, page 116

B Vocabulary in use

CORE TERMS

user need
A requirement expressed from the perspective of the intended user.

capture a user need

design input
A specified requirement that guides device design and development.

define a design input

design output
A result of design work that can be checked against inputs.

review a design output

intended user
The person or user group the device is designed for.

identify intended users

intended use
The purpose for which a device is intended.

define the intended use

use environment
The physical and social setting in which a device is used.

characterize the use environment

user profile
Relevant characteristics of a defined user group.

develop a user profile

task analysis
Examination of the actions and decisions needed to perform a task.

conduct a task analysis

use scenario
A description of users performing tasks under specified conditions.

map a use scenario

design constraint
A limit that the design must satisfy.

document a design constraint

acceptance criterion
A specified condition used to decide whether a requirement is met.

establish acceptance criteria

measurable requirement
A requirement expressed so its fulfillment can be evaluated.

formulate a measurable requirement

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

user need USER NEED

capture a user need

Keep the meaning precise: A requirement expressed from the perspective of the intended user.

use environment USE ENVIRONMENT

characterize the use environment

Keep the meaning precise: The physical and social setting in which a device is used.

B Precision and professional shorthand

EXTEND YOUR RANGE

stakeholder

A person or group with a relevant interest in the product.

consult stakeholders

voice of the customer

Collected customer needs and expectations.

analyze the voice of the customer

requirements baseline

The agreed requirement set used as a reference for controlled work.

approve a requirements baseline

traceability

The ability to follow links between related records or requirements.

maintain requirements traceability

use limitation

A boundary on the conditions or purposes of use.

state a use limitation

ergonomics

Design attention to how people physically and cognitively interact with a product.

evaluate ergonomics

interface

The elements through which a user interacts with a device.

assess the user interface

workflow

The ordered activities through which work is completed.

observe the clinical workflow

design assumption

An unconfirmed premise influencing a design decision.

challenge a design assumption

scope expansion

An increase in the users, uses, or conditions covered by a project.

assess a scope expansion

change request

A documented proposal to alter an agreed item.

raise a change request

requirement owner

The person accountable for clarifying and maintaining a requirement.

assign a requirement owner

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

measurable requirement MEASURABLE REQUIREMENT

formulate a measurable requirement

Keep the meaning precise: A requirement expressed so its fulfillment can be evaluated.

use limitation USE LIMITATION

state a use limitation

Keep the meaning precise: A boundary on the conditions or purposes of use.

C Phrases that move the work forward

Acknowledge the need

Ease of use matters; we need to define the task behind that phrase.

Separate an expansion

Home caregivers are a proposed additional user group.

Clarify the task

What exactly counts as setup in this requirement?

Ask for the basis

What evidence supports that time limit?

Request traceability

Link this input to the need and the planned evaluation.

Identify an assumption

We are assuming equivalent training, but that has not been established.

Request a review

The scope change needs review before it enters the baseline.

Preserve status

Keep the home-use item marked proposed in the meeting record.

Name the current audience

The agreed initial users are trained clinic staff.

Ask about the setting

Which lighting, noise, and workflow conditions are in scope?

Mark a proposal

Two minutes is a candidate target, not an approved criterion.

Keep needs visible

The requirement should address the user need rather than only our preferred feature.

Challenge a broad claim

Suitable everywhere goes beyond the environments we have assessed.

Define ownership

Who will resolve the user-profile question?

Distinguish output

A new screen layout is an output, not proof that the need is met.

Close the handoff

We will return with the defined task, conditions, and criterion basis.

Precision in real exchanges

Purpose: Ask about the setting
Useful wording: Which lighting, noise, and workflow conditions are in scope?

Purpose: Clarify the task
Useful wording: What exactly counts as setup in this requirement?

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which revision preserves the current scope?

A. Delete every reference to users and retain easy to use.

B. Design for trained clinic staff; assess home-caregiver use as a proposed expansion.

C. Design for all users because the interface is unchanged.

D. State home use is validated because it appears on a slide.

2. Which statement accurately describes the time target?

A. Two minutes is proposed; task boundaries and the criterion basis need review.

B. Two minutes is approved because it is measurable.

C. A numeric target needs no justification.

D. The target proves suitability for home use.

3. A colleague needs the term for this meaning: The person or user group the device is designed for. Which term fits?

A. design assumption

B. intended user

C. use scenario

D. requirements baseline

D Choose the response that fits / continued

4. Rebuild the exchange from "What does easy to use mean?". Owen: "Could we write setup within two minutes? It sounds clearer than easy to use, and the number would give engineering something definite to work toward." Mira: [missing response] Owen: "That distinction matters. I do not want the meeting notes to turn a suggested number into a promise that the product has already been shown to meet." Which response belongs in the gap?

A. We should also characterize the use environment. Lighting, interruptions, noise, and access to other staff can affect the task even when the screen stays unchanged.

B. Treat the screen layout as a design output. The input should express what must be achieved without presenting an untested layout as proof of success.

C. We can retain it as a user need. We still need the task, the people performing it, and the conditions that make the task difficult.

D. It is a candidate acceptance criterion, not an approved one. We must define what setup includes and why two minutes is appropriate for those users.

Answers and explanations, page 116

One useful distinction

A proposed two-minute target is not an approved acceptance criterion. The team must identify what setup includes, who performs it, under which conditions, and why the target is appropriate before treating the number as a settled requirement.

E Conversation

What does easy to use mean?

What does easy to use mean?

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

user need / user profile / design input / design output / requirements baseline / scope expansion / traceability / use environment / acceptance criterion / requirement owner

01 **Owen:** The customer summary says the device should be easy to use. I agree with that direction, but the design team says it cannot work from that sentence alone.

02 **Mira:** We can retain it as a [1] _____. We still need the task, the people performing it, and the conditions that make the task difficult.

03 **Owen:** The initial audience is trained clinic staff. The latest slide adds home caregivers because sales sees a possible opportunity, although that group has not been assessed.

04 **Mira:** Then mark that as a [2] _____. It is not simply another way of describing the audience already agreed in the brief.

05 **Owen:** Could we write setup within two minutes? It sounds clearer than easy to use, and the number would give engineering something definite to work toward.

06 **Mira:** It is a candidate [3] _____, not an approved one. We must define what setup includes and why two minutes is appropriate for those users.

07 **Owen:** That distinction matters. I do not want the meeting notes to turn a suggested number into a promise that the product has already been shown to meet.

08 **Mira:** We should also characterize the [4] _____. Lighting, interruptions, noise, and access to other staff can affect the task even when the screen stays unchanged.

09 **Owen:** The sales slide says suitable everywhere. We have not gathered evidence from homes, so I will remove that claim from the current product description.

10 **Mira:** Good. The [5] _____ should describe relevant training and capabilities. We cannot assume that clinic staff and home caregivers approach the same instruction in the same way.

Conversation continues on the next page.

What does easy to use mean?

CONVERSATION / CONTINUED

11 **Owen:** Engineering wants a simpler screen. Is that the requirement, or their proposed way to meet it? I want the brief to distinguish the two.

12 **Mira:** Treat the screen layout as a [6] _____. The input should express what must be achieved without presenting an untested layout as proof of success.

13 **Owen:** Then we need a record that connects the customer concern to a reviewable input and eventually to evidence. Otherwise the same question will return at every review.

14 **Mira:** That is [7] _____. Each link should make the reasoning visible, including which assumptions remain open and which version the team has agreed.

15 **Owen:** Who should resolve the setup definition? I can bring customer context, but I should not independently decide the engineering criterion or the relevant evaluation conditions.

16 **Mira:** Assign a [8] _____ to coordinate those contributions. The owner can bring the proposed wording and its basis back for the appropriate review.

17 **Owen:** I will separate the clinic requirement from the home-use proposal. The notes will say that the two-minute target remains unreviewed, with a named follow-up owner.

18 **Mira:** Keep both proposals outside the approved [9] _____ until the review is complete. Clear status now prevents later teams from treating a meeting idea as an established obligation.

19 **Owen:** I will keep easy to use as the customer concern and assign the setup-definition work. The home-use proposal and two-minute target remain open for review.

20 **Mira:** Exactly. A well-scoped [10] _____ will identify what the design must achieve. The evidence must then show whether the resulting device actually meets the relevant requirement.

CHECK THE GAPS

1. user need / **2.** scope expansion / **3.** acceptance criterion / **4.** use environment / **5.** user profile / **6.** design output / **7.** traceability / **8.** requirement owner / **9.** requirements baseline / **10.** design input

Why these answers fit, page 117

F Check accuracy and scope

What does easy to use mean?

Return to the case

Design engineer Mira and product lead Owen review a fictional monitoring-device brief. It says easy to use and suitable everywhere. The agreed initial audience is trained clinic staff, but an appended slide also mentions home caregivers. No home-use assessment is supplied. A proposed setup-time target of two minutes has not been reviewed or approved. Mira and Owen must distinguish the agreed scope, the proposed expansion, and the missing basis for a measurable design input.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. What best distinguishes a need from an output?

A. The need and the layout are identical records.

B. Any output proves the need is satisfied.

C. A user need must specify a preferred screen color.

D. The need describes the user requirement; the screen layout is a design result.

2. Which meeting record is most accurate?

A. Home use rejected permanently by this discussion.

B. Setup performance failed a completed validation study.

C. Clinic scope agreed; home-use assessment and setup criterion remain open.

D. All environments approved and two-minute performance established.

Answers and explanations, page 119

Listen for the distinction

Swap roles. State the two-minute target as unapproved; identify setup boundaries and the basis still needed.

Clarify without dismissing the need

A customer adjective can be a useful starting point. Acknowledge the concern behind it, then request the user, task, conditions, and evaluation basis. This sounds more collaborative than rejecting the adjective as meaningless, while still preventing an unsupported requirement from becoming an accepted fact.

F Say it / transfer to a new case

A new setting, an open requirement

A brief covers trained outpatient staff. A slide proposes home use. A ninety-second task target is suggested but has not been approved.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

criterion / clinic / expansion / baseline

01 **Lead:** The agreed setting is the outpatient [1] _____.

02 **Engineer:** Home use is a proposed [2] _____.

03 **Lead:** Ninety seconds is not yet an approved [3] _____.

04 **Engineer:** Keep the proposal separate from the current [4] _____.

CHECK THE GAPS

1. clinic / **2.** expansion / **3.** criterion / **4.** baseline

Why these answers fit, page 119

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "What does easy to use mean?". Owen: "Then we need a record that connects the customer concern to a reviewable input and eventually to evidence. Otherwise the same question will return at every review." Mira: [missing response] Owen: "Who should resolve the setup definition? I can bring customer context, but I should not independently decide the engineering criterion or the relevant evaluation conditions." Which response belongs in the gap?

A. Then mark that as a scope expansion. It is not simply another way of describing the audience already agreed in the brief.

B. We can retain it as a user need. We still need the task, the people performing it, and the conditions that make the task difficult.

C. Assign a requirement owner to coordinate those contributions. The owner can bring the proposed wording and its basis back for the appropriate review.

D. That is traceability. Each link should make the reasoning visible, including which assumptions remain open and which version the team has agreed.

Answers and explanations, page 120

A Read the situation

LESSON 02 / RISK MANAGEMENT AND HAZARD ANALYSIS

Separate a hazard, an exposure situation, possible harm, and the evidence for a risk-control claim.

The hot surface is not the injury

Risk engineer Elena and mechanical lead Sam review a fictional enclosure-heating concern. The worksheet labels skin burn as the hazard and says low risk because no complaints have been received. The team has identified a potentially hot accessible surface, possible prolonged contact, and possible skin injury. No likelihood estimate or verified control result is supplied. An insulating cover is only a proposal. Their task is to correct the chain of reasoning without deciding that the device is safe.

People: Elena, Risk engineer; Sam, Mechanical lead.

Read the record

Choose one answer per question using the facts above.

1. Which item is the potential source of harm?

A. The absence of complaints
B. The unapproved meeting minutes
C. The accessible hot surface
D. The possible skin burn

2. What is the status of the insulating cover?

A. Formally accepted as eliminating every risk
B. A confirmed cause of the original heating
C. Proposed, with no verified control result supplied
D. Implemented and proven effective

3. Why is low risk not established?

A. Possible harm and likelihood are identical concepts.
B. Neither a likelihood estimate nor a verified control result is supplied.
C. Every hot surface necessarily causes injury.
D. Any complaint-free device is automatically safe.

Answers and explanations, page 121

B Vocabulary in use

CORE TERMS

hazard
A potential source of injury or other harm.

identify a hazard

harm
Injury or damage to health, property, or the environment.

describe possible harm

hazardous situation
Circumstances in which people or other protected things are exposed to a hazard.

identify a hazardous situation

sequence of events
The linked events that can lead to exposure and harm.

map the sequence of events

severity
The degree of possible harm.

assess harm severity

probability of harm
The likelihood that the specified harm will occur.

estimate the probability of harm

risk
The combination of the probability of harm and its severity.

evaluate a risk

risk analysis
Systematic identification of hazards and estimation of associated risks.

document the risk analysis

risk evaluation
Comparing estimated risk with established acceptability criteria.

perform a risk evaluation

risk control
A measure intended to reduce risk.

implement a risk control

residual risk
Risk remaining after risk-control measures have been implemented.

evaluate residual risk

risk acceptability
Whether risk meets the applicable acceptance criteria.

document risk acceptability

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

hazard HAZARD

identify a hazard

Keep the meaning precise: A potential source of injury or other harm.

probability of harm PROBABILITY OF HARM

estimate the probability of harm

Keep the meaning precise: The likelihood that the specified harm will occur.

B Precision and professional shorthand

EXTEND YOUR RANGE

protective measure
A safeguard intended to reduce exposure or its consequences.

assess a protective measure

inherent safety
Safety achieved through the design itself.

consider inherently safe design

information for safety
Warnings or instructions intended to support risk reduction.

review information for safety

benefit-risk analysis
Evaluation of relevant benefits against remaining risks.

document a benefit-risk analysis

risk management file
The organized records of risk-management activities and results.

maintain the risk management file

foreseeable misuse
Incorrect use that can reasonably be anticipated.

assess foreseeable misuse

normal condition
Operation without a fault being present.

assess normal conditions

fault condition
A state in which a component or function has failed.

examine fault conditions

control effectiveness
The demonstrated ability of a measure to reduce the identified risk.

verify control effectiveness

newly introduced risk
A risk created or changed by a design or control modification.

assess newly introduced risks

risk estimate
An assessment of harm probability and severity.

justify a risk estimate

objective evidence
Verifiable information supporting a conclusion.

obtain objective evidence

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

risk acceptability RISK ACCEPTABILITY

document risk acceptability

Keep the meaning precise: Whether risk meets the applicable acceptance criteria.

risk management file RISK MANAGEMENT FILE

maintain the risk management file

Keep the meaning precise: The organized records of risk-management activities and results.

C Phrases that move the work forward

Identify the source

The accessible hot surface is the potential source of harm.

Name the exposure

Prolonged contact is the exposure situation we need to examine.

Describe the outcome

A skin burn is the possible harm, not the source.

Separate two dimensions

Severity and probability answer different questions.

Ask for the basis

What evidence supports this likelihood estimate?

Avoid false reassurance

No complaints received does not establish zero risk.

Mark a proposed control

The cover is a candidate protective measure.

Request evidence

We have not yet verified the control's effectiveness.

Check the chain

Which sequence could lead from this condition to injury?

Preserve uncertainty

The current record does not support the low-risk conclusion.

Assess the change

Could the proposed cover introduce another risk?

Clarify residual risk

We cannot describe post-control risk before assessing the implemented measure.

Name the criterion

Which acceptability criteria are being applied?

Keep the record linked

Connect the analysis, control, verification, and review decision.

Avoid a guarantee

This review does not establish that every risk has been eliminated.

Close with ownership

Assign the missing evidence to an owner before the next review.

Precision in real exchanges

Purpose: Separate two dimensions
Useful wording: Severity and probability answer different questions.

Purpose: Ask for the basis
Useful wording: What evidence supports this likelihood estimate?

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which chain is correctly labeled?

A. Burn: hazard; hot surface: harm; no complaints: control.

B. Prolonged contact: proof of injury; burn: probability.

C. Cover proposal: verified control; burn: acceptance criterion.

D. Hot surface: hazard; prolonged contact: exposure situation; burn: harm.

2. Which sentence accurately reports the complaint history?

A. No complaints prove that contact cannot occur.

B. No complaints prove that the cover works.

C. No complaints make severity irrelevant.

D. No complaints have been received; this alone does not establish the likelihood of harm.

3. A colleague needs the term for this meaning: The linked events that can lead to exposure and harm. Which term fits?

A. control effectiveness

B. risk evaluation

C. sequence of events

D. information for safety

D Choose the response that fits / continued

4. Rebuild the exchange from "The hot surface is not the injury". Sam: "Then the burn belongs in the outcome field. We have not established that it happened, so the wording must not imply an observed patient event." Elena: [missing response] Sam: "The team chose low because the complaint log is empty. That seems reassuring, but I cannot find any estimate showing how the exposure might occur." Which response belongs in the gap?

A. Keep severity separate from likelihood. We need to describe the possible injury carefully; uncertainty in one dimension does not make the other dimension disappear.

B. Correct. It is possible harm. State the outcome conditionally and preserve the distinction between this design concern and a documented incident.

C. Yes, document the applicable risk acceptability criteria. The reviewer needs the reasoning and evidence, not simply a low label attached to an unfinished analysis.

D. Not from this record. The accessible hot surface is the hazard. A skin burn describes a possible outcome, and the cover has not been evaluated.

Answers and explanations, page 121

One useful distinction

Hazard, hazardous situation, and harm are not interchangeable. In this case, the potentially hot surface is the source; prolonged contact describes exposure; a burn is the possible injury. These distinctions clarify the analysis without supplying a likelihood estimate.

E Conversation

The hot surface is not the injury

The hot surface is not the injury

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

newly introduced risk / hazard / harm / risk control / severity / probability of harm / Residual risk / risk acceptability / hazardous situation / objective evidence

01 **Sam:** The worksheet calls the skin burn a hazard and rates the issue low because we have no complaints. Can we close it with the proposed insulating cover?

02 **Elena:** Not from this record. The accessible hot surface is the [1] _____. A skin burn describes a possible outcome, and the cover has not been evaluated.

03 **Sam:** I see the labeling problem. The surface could be accessible during use, and the scenario includes prolonged contact. We should keep those details in separate fields.

04 **Elena:** Yes. That contact describes a [2] _____. It helps us explain how someone could be exposed instead of jumping directly from a component to an injury.

05 **Sam:** Then the burn belongs in the outcome field. We have not established that it happened, so the wording must not imply an observed patient event.

06 **Elena:** Correct. It is possible [3] _____. State the outcome conditionally and preserve the distinction between this design concern and a documented incident.

07 **Sam:** The team chose low because the complaint log is empty. That seems reassuring, but I cannot find any estimate showing how the exposure might occur.

08 **Elena:** We still need a justified [4] _____. An empty log does not establish zero occurrence, especially without understanding exposure, reporting, and the underlying evidence.

09 **Sam:** Should we remove the severity entry as well? It seems difficult to judge the risk while the likelihood information remains incomplete in this version.

10 **Elena:** Keep [5] _____ separate from likelihood. We need to describe the possible injury carefully; uncertainty in one dimension does not make the other dimension disappear.

Conversation continues on the next page.

The hot surface is not the injury

CONVERSATION / CONTINUED

11 **Sam:** The cover remains attractive as an engineering proposal. I want to record it without giving the impression that it has already solved the issue.

12 **Elena:** Call it a proposed [6] _____. Identify what it is intended to change, the evidence needed, and who will review whether that change is effective.

13 **Sam:** Could the cover trap heat somewhere else? I want the review to consider that possibility as well as reducing contact with the hot surface.

14 **Elena:** Exactly. Assess any [7] _____. A measure can address one exposure while affecting another part of the design, so its consequences need a broader review.

15 **Sam:** The draft already says residual risk acceptable. That sounds premature when the proposed measure has not been implemented or assessed against the relevant criteria.

16 **Elena:** It is premature. [8] _____ refers to what remains after controls. We should not describe a future condition as though its implementation and effectiveness were already established.

17 **Sam:** I will change the status to evidence incomplete and retain the proposed cover as an action for engineering review. We also need the criterion behind any acceptance decision.

18 **Elena:** Yes, document the applicable [9] _____ criteria. The reviewer needs the reasoning and evidence, not simply a low label attached to an unfinished analysis.

19 **Sam:** I will separate the hot surface, prolonged contact, and possible burn in the record. Send me the evidence needed before we can judge the proposed cover.

20 **Elena:** That gives the next review useful [10] _____ to request and evaluate. We can make progress on the concern without converting a proposal into a completed safety conclusion.

CHECK THE GAPS

1. hazard / **2.** hazardous situation / **3.** harm / **4.** probability of harm / **5.** severity / **6.** risk control / **7.** newly introduced risk / **8.** Residual risk / **9.** risk acceptability / **10.** objective evidence

Why these answers fit, page 122

F Check accuracy and scope

The hot surface is not the injury

Return to the case

Risk engineer Elena and mechanical lead Sam review a fictional enclosure-heating concern. The worksheet labels skin burn as the hazard and says low risk because no complaints have been received. The team has identified a potentially hot accessible surface, possible prolonged contact, and possible skin injury. No likelihood estimate or verified control result is supplied. An insulating cover is only a proposal. Their task is to correct the chain of reasoning without deciding that the device is safe.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which review would support a control claim for the proposed insulating cover?

A. Its proposed geometry, with residual risk accepted before implementation
B. Its supplier description, with complaint absence substituted for control testing
C. Its implementation, effect on contact risk, and any risk introduced by the change
D. Its presence in the drawing, with effectiveness inferred from its protective name

2. Which conclusion fits the case?

A. The device is proven safe for every use.
B. A burn certainly occurred in every unit.
C. The cover has already eliminated the hazard.
D. Further evidence is needed before the stated risk conclusion is supported.

Answers and explanations, page 124

Listen for the distinction

Swap roles. Keep the cover proposed and its effect unverified; do not use complaint absence as a probability estimate.

Challenge the chain, not the person

Risk meetings become more productive when colleagues identify the exact missing link. Ask whether a phrase names a source, an exposure, an outcome, or a control. A neutral classification question can reveal an important disagreement without turning the review into a personal argument.

F Say it / transfer to a new case

Proposal is not verification

A fictional handle may have a sharp edge. Hand contact could cause a cut. A rounded redesign is proposed but has not been evaluated.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

hazard / control / evidence / harm

01 **Engineer:** The sharp edge is the potential [1] _____.

02 **Reviewer:** The cut is the possible [2] _____.

03 **Engineer:** The rounded redesign is a proposed [3] _____.

04 **Reviewer:** Its effectiveness still requires [4] _____.

CHECK THE GAPS

1. hazard / **2.** harm / **3.** control / **4.** evidence

Why these answers fit, page 124

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "The hot surface is not the injury". Sam: "Could the cover trap heat somewhere else? I want the review to consider that possibility as well as reducing contact with the hot surface." Elena: [missing response] Sam: "The draft already says residual risk acceptable. That sounds premature when the proposed measure has not been implemented or assessed against the relevant criteria." Which response belongs in the gap?

A. Exactly. Assess any newly introduced risk. A measure can address one exposure while affecting another part of the design, so its consequences need a broader review.

B. We still need a justified probability of harm. An empty log does not establish zero occurrence, especially without understanding exposure, reporting, and the underlying evidence.

C. Yes. That contact describes a hazardous situation. It helps us explain how someone could be exposed instead of jumping directly from a component to an injury.

D. Not from this record. The accessible hot surface is the hazard. A skin burn describes a possible outcome, and the cover has not been evaluated.

Answers and explanations, page 125

Review 1 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "User Needs and Design Inputs" and "Risk Management and Hazard Analysis". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A result of design work that can be checked against inputs. Which term fits?

A. design output
B. voice of the customer
C. task analysis
D. workflow

2. A colleague needs the term for this meaning: Systematic identification of hazards and estimation of associated risks. Which term fits?

A. risk analysis
B. protective measure
C. hazard
D. normal condition

3. Rebuild the exchange from "What does easy to use mean?". Owen: "The sales slide says suitable everywhere. We have not gathered evidence from homes, so I will remove that claim from the current product description." Mira: [missing response] Owen: "Engineering wants a simpler screen. Is that the requirement, or their proposed way to meet it? I want the brief to distinguish the two." Which response belongs in the gap?

A. Good. The user profile should describe relevant training and capabilities. We cannot assume that clinic staff and home caregivers approach the same instruction in the same way.
B. We can retain it as a user need. We still need the task, the people performing it, and the conditions that make the task difficult.
C. Exactly. A well-scoped design input will identify what the design must achieve. The evidence must then show whether the resulting device actually meets the relevant requirement.
D. Keep both proposals outside the approved requirements baseline until the review is complete. Clear status now prevents later teams from treating a meeting idea as an established obligation.

Review 1 / Meaning and context / continued

4. Rebuild the exchange from "The hot surface is not the injury". Sam: "I will change the status to evidence incomplete and retain the proposed cover as an action for engineering review. We also need the criterion behind any acceptance decision." Elena: [missing response] Sam: "I will separate the hot surface, prolonged contact, and possible burn in the record. Send me the evidence needed before we can judge the proposed cover." Which response belongs in the gap?

A. We still need a justified probability of harm. An empty log does not establish zero occurrence, especially without understanding exposure, reporting, and the underlying evidence.

B. Keep severity separate from likelihood. We need to describe the possible injury carefully; uncertainty in one dimension does not make the other dimension disappear.

C. Yes, document the applicable risk acceptability criteria. The reviewer needs the reasoning and evidence, not simply a low label attached to an unfinished analysis.

D. Not from this record. The accessible hot surface is the hazard. A skin burn describes a possible outcome, and the cover has not been evaluated.

Answers and explanations, page 156

A Read the situation

LESSON 03 / VERIFICATION, VALIDATION, AND DESIGN REVIEW

Report completed specification checks without implying that intended-use evaluation or release authorization is complete.

> **Twelve checks passed, one evaluation still open**
>
> Verification lead Arun and development manager Beth review the status of a fictional device. All twelve planned specification checks in the current package passed. The separate intended-user evaluation has not started. The slide nevertheless says validation complete and ready for release. No release authorization is supplied. A review meeting is scheduled tomorrow. Arun and Beth must present the completed evidence, the unperformed work, and the decision still required without treating the meeting itself as approval.

People: Arun, Verification lead; Beth, Development manager.

Read the record

Choose one answer per question using the facts above.

1. What has passed?

A. The final release authorization
B. All future product configurations
C. Twelve planned specification checks in the current package
D. Every intended-user evaluation

2. What has not started?

A. The scheduled review meeting yesterday
B. The separate intended-user evaluation
C. Every specification check
D. The project itself

3. What does tomorrow's meeting establish now?

A. That all open actions are closed
B. Only that a review is scheduled
C. That release is authorized
D. That intended use has been validated

Answers and explanations, page 126

B Vocabulary in use

CORE TERMS

verification
Confirmation with evidence that specified requirements have been fulfilled.

complete design verification

validation
Confirmation with evidence that requirements for an intended use are fulfilled.

plan design validation

design review
A systematic evaluation of design progress, evidence, and issues.

hold a design review

test protocol
A documented plan describing how a test will be performed and assessed.

approve a test protocol

test report
A record of test execution, results, and conclusions.

review a test report

test article
The item or configuration used in a test.

identify the test article

configuration
The defined arrangement and versions of a product's elements.

control the test configuration

requirement coverage
The extent to which requirements are addressed by specified evidence.

assess requirement coverage

traceability matrix
A structured table linking requirements to related evidence.

update the traceability matrix

test deviation
A departure from the planned test method or conditions.

assess a test deviation

unresolved anomaly
An observed irregularity whose significance or resolution remains open.

review unresolved anomalies

acceptance result
The recorded outcome against a specified acceptance criterion.

document the acceptance result

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

verification VERIFICATION

complete design verification

Keep the meaning precise: Confirmation with evidence that specified requirements have been fulfilled.

test article TEST ARTICLE

identify the test article

Keep the meaning precise: The item or configuration used in a test.

B Precision and professional shorthand

EXTEND YOUR RANGE

representativeness

How well an item or setting reflects the intended evaluated population or use.

justify representativeness

simulated use

Evaluation under conditions designed to represent actual use.

define simulated-use conditions

review action

A task assigned as a result of a review.

track review actions

action closure

Documented completion and acceptance of an assigned task.

verify action closure

release authorization

A formal decision permitting a product or version to be released.

obtain release authorization

design transfer

Translation of the design into production specifications and processes.

review design transfer

document control

Management of document versions, approval, and availability.

maintain document control

change impact

The consequences of a proposed or implemented alteration.

assess change impact

regression testing

Rechecking affected behavior after a change.

plan regression testing

evidence package

The organized records supporting a specific conclusion or decision.

assemble the evidence package

review quorum

The required participation for a particular review under its procedure.

confirm the review quorum

decision record

A documented conclusion, rationale, and associated conditions.

maintain a decision record

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

acceptance result ACCEPTANCE RESULT

document the acceptance result

Keep the meaning precise: The recorded outcome against a specified acceptance criterion.

release authorization RELEASE AUTHORIZATION

obtain release authorization

Keep the meaning precise: A formal decision permitting a product or version to be released.

C Phrases that move the work forward

State the completed work

All twelve checks in this specification package passed.

Name the open evaluation

The intended-user evaluation has not started.

Request the link

Which requirement does this result address?

Separate the decision

A passed test package is not release authorization.

Keep actions visible

Assign an owner and closure evidence for each open item.

Request the rationale

What supports the representativeness of this test article?

Avoid a global claim

We have not demonstrated every intended-use requirement.

Record the conclusion

Capture the actual decision and any conditions after the review.

Limit the conclusion

That result applies to the tested configuration and covered requirements.

Correct the status

Replace validation complete with specification checks complete.

Ask about deviations

Were any departures from the protocol assessed and documented?

Describe the meeting

The review is scheduled; its conclusion is not yet available.

Check the version

Confirm that the report covers the version presented for review.

Assess a change

A later modification needs an impact assessment before relying on these results.

Clarify the handoff

The review package should show completed evidence and unresolved work separately.

Close accurately

Ready for review does not mean ready for release.

Precision in real exchanges

Purpose: Correct the status
Useful wording: Replace validation complete with specification checks complete.

Purpose: Request the link
Useful wording: Which requirement does this result address?

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which slide heading is supported?

A. Twelve specification checks passed; intended-user evaluation not started.

B. Validation complete and product released.

C. All user needs proven by twelve engineering checks.

D. Review completed because tomorrow is booked.

2. A software revision follows the tests. What wording is best?

A. Any change makes all historical evidence false.

B. A new version number alone proves validation.

C. Assess the change before assuming the earlier results cover the new version.

D. Earlier results automatically cover every future revision.

3. A colleague needs the term for this meaning: A documented plan describing how a test will be performed and assessed. Which term fits?

A. traceability matrix

B. test protocol

C. regression testing

D. review action

D Choose the response that fits / continued

4. Rebuild the exchange from "Twelve checks passed, one evaluation still open". Beth: "Can we say every requirement passed? That would be shorter, although I suspect the word every would extend the claim beyond this particular set of checks." Arun: [missing response] Beth: "I also noticed that the report refers to the earlier software build. The team discussed a revision, but I do not know whether it has been implemented." Which response belongs in the gap?

A. Show the requirement coverage instead. Each result needs a clear link to its requirement, and any uncovered requirements must remain visible in the review package.

B. The completed work is verification against the specifications in this package. The separate intended-user evaluation has not started, so the headline combines two different claims.

C. Exactly. We cannot mark validation complete from these checks alone. The report should distinguish specification fulfillment from the evidence still needed about the intended use.

D. Yes. Action closure needs the required evidence and acceptance. Ownership makes follow-up possible; it does not demonstrate that the remaining task has already been completed.

Answers and explanations, page 126

> **One useful distinction**
>
> The twelve checks support the specified requirements they address, in the tested configuration. They do not automatically cover every user need, later version, or intended-use condition. Report completion at the level supported by the actual evidence.

E Conversation

Twelve checks passed, one evaluation still open

Twelve checks passed, one evaluation still open

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

verification / Release authorization / decision record / configuration / validation / Action closure / test deviation / change impact / evidence package / requirement coverage

01 **Beth:** The dashboard says validation complete because all twelve checks passed. Before tomorrow's review, I want to be sure that headline means what the team thinks it means.

02 **Arun:** The completed work is [1] _____ against the specifications in this package. The separate intended-user evaluation has not started, so the headline combines two different claims.

03 **Beth:** The test result is still good progress. We should not hide it, but we should identify what those twelve checks actually cover and what remains unperformed.

04 **Arun:** Exactly. We cannot mark [2] _____ complete from these checks alone. The report should distinguish specification fulfillment from the evidence still needed about the intended use.

05 **Beth:** Can we say every requirement passed? That would be shorter, although I suspect the word every would extend the claim beyond this particular set of checks.

06 **Arun:** Show the [3] _____ instead. Each result needs a clear link to its requirement, and any uncovered requirements must remain visible in the review package.

E Conversation / continued

07 **Beth:** I also noticed that the report refers to the earlier software build. The team discussed a revision, but I do not know whether it has been implemented.

08 **Arun:** We must confirm the [4] _____. Evidence belongs to the item actually tested; a discussion about a later build cannot silently extend the report to that version.

09 **Beth:** If the revision goes ahead, which test conclusions might it affect? Please flag that question now so the release review does not inherit mismatched versions.

10 **Arun:** That calls for a [5] _____ assessment. We should not assume that every old result becomes invalid, or that every result automatically remains sufficient.

Conversation continues on the next page.

Twelve checks passed, one evaluation still open

CONVERSATION / CONTINUED

11 **Beth:** The dashboard also says ready for release. I intended ready for tomorrow's meeting, but I can see that those are not interchangeable status descriptions.

12 **Arun:** Correct. [6] _____ is a separate decision. We have not been given that authorization, and scheduling a review does not establish its future outcome.

13 **Beth:** I will change the status to ready for review and put the unperformed intended-user evaluation beside the completed specification package. That should prevent a misleading green summary.

14 **Arun:** Include the [7] _____, not just the totals. Reviewers need the reports, relevant versions, criteria, and any departures from the planned method.

15 **Beth:** There was one departure noted during execution. I do not want to describe it as a failed requirement if that is not what happened.

16 **Arun:** Call it a [8] _____ and reference its assessment. The record should explain its significance and disposition without treating the label itself as a pass or failure.

17 **Beth:** Tomorrow, each unresolved item needs an owner. But the minutes should not show those items as closed simply because someone agrees to take responsibility.

18 **Arun:** Yes. [9] _____ needs the required evidence and acceptance. Ownership makes follow-up possible; it does not demonstrate that the remaining task has already been completed.

19 **Beth:** I will separate the twelve passed checks from the unstarted user evaluation. Please confirm the tested build and keep release authorization off the slide until it exists.

20 **Arun:** After the meeting, add the actual [10] _____ and its conditions. That keeps completed testing, open evaluation, review conclusions, and release status distinct throughout the handoff.

CHECK THE GAPS

1. verification / **2.** validation / **3.** requirement coverage / **4.** configuration / **5.** change impact / **6.** Release authorization / **7.** evidence package / **8.** test deviation / **9.** Action closure / **10.** decision record

Why these answers fit, page 127

F Check accuracy and scope

Twelve checks passed, one evaluation still open

Return to the case

Verification lead Arun and development manager Beth review the status of a fictional device. All twelve planned specification checks in the current package passed. The separate intended-user evaluation has not started. The slide nevertheless says validation complete and ready for release. No release authorization is supplied. A review meeting is scheduled tomorrow. Arun and Beth must present the completed evidence, the unperformed work, and the decision still required without treating the meeting itself as approval.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. A reviewer accepts ownership of an open test deviation. Which status can be recorded at that point?

A. Release approved because the deviation has entered the action log

B. Owner assigned; required assessment, completion evidence, and acceptance still outstanding

C. Deviation accepted as harmless because a reviewer now owns it

D. Action closed because responsibility is no longer unassigned

2. Which conclusion may Beth report before the review?

A. The review has approved release.

B. Intended-use validation succeeded without being performed.

C. No further decision is required.

D. The evidence package is ready for the scheduled review, with the remaining work identified.

Answers and explanations, page 129

Listen for the distinction

Swap roles. Distinguish the tested configuration, tomorrow's review, and the release decision still unmade.

Use complete with an object

A status update can be confident and narrow at the same time. Say which package is complete and which criterion it addresses. Replacing a global green status with separate evidence and decision lines helps colleagues see progress without confusing a finished test with a finished product.

F Say it / transfer to a new case

A review date is not a decision

Six specification checks passed on version B. The intended-use evaluation is pending, and a review is booked for Friday. No release decision has been made.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

authorized / review / B / pending

01 **Lead:** Six checks passed on version [1] _____.

02 **Engineer:** The intended-use evaluation remains [2] _____.

03 **Lead:** Friday is the scheduled [3] _____.

04 **Engineer:** Release has not been [4] _____.

CHECK THE GAPS

1. B / **2.** pending / **3.** review / **4.** authorized

Why these answers fit, page 129

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Twelve checks passed, one evaluation still open". Beth: "I will change the status to ready for review and put the unperformed intended-user evaluation beside the completed specification package. That should prevent a misleading green summary." Arun: [missing response] Beth: "There was one departure noted during execution. I do not want to describe it as a failed requirement if that is not what happened." Which response belongs in the gap?

A. Correct. Release authorization is a separate decision. We have not been given that authorization, and scheduling a review does not establish its future outcome.

B. Include the evidence package, not just the totals. Reviewers need the reports, relevant versions, criteria, and any departures from the planned method.

C. That calls for a change impact assessment. We should not assume that every old result becomes invalid, or that every result automatically remains sufficient.

D. The completed work is verification against the specifications in this package. The separate intended-user evaluation has not started, so the headline combines two different claims.

Answers and explanations, page 130

A Read the situation

LESSON 04 / USABILITY AND HUMAN FACTORS

Report observed use difficulties neutrally and distinguish exploratory findings from validation or population-level rates.

Three participants read the label differently

Human-factors specialist Laila and interface designer Victor review a formative study of a fictional device label. Three of eight participants selected the wrong status label during a simulated task. The draft summary calls them careless and concludes that a larger font will solve the problem. No revised label has been tested, and the study does not establish a population error rate. Laila and Victor must preserve the observation, investigate the interface, and describe the proposed revision without declaring success.

People: Laila, Human-factors specialist; Victor, Interface designer.

Read the record

Choose one answer per question using the facts above.

1. What was observed?

A. Every user failed in actual clinical use.
B. A larger font eliminated the difficulty.
C. A population-wide error rate was established.
D. Three of eight participants selected the wrong status label in the simulated task.

2. Which wording adds a cause not established by the three wrong selections?

A. Three participants selected the wrong status label in the task.
B. The proposed larger-font version has not been evaluated.
C. The reason for the selections remains under investigation.
D. Participants selected incorrectly because they were careless.

3. What is the status of the larger font?

A. An untested proposed revision
B. A validated solution
C. An approved universal corrective action
D. Proof that no other interface issue exists

Answers and explanations, page 131

B Vocabulary in use

CORE TERMS

human factors
Study and application of how people interact with systems and products.

apply human-factors principles

usability engineering
A structured approach to evaluating and improving user interaction.

conduct usability engineering

formative evaluation
An exploratory assessment used to inform design development.

run a formative evaluation

summative evaluation
An assessment of a sufficiently developed design against defined evaluation goals.

plan a summative evaluation

human-factors validation
Evaluation of intended users performing relevant tasks under representative conditions.

plan human-factors validation

critical task
A task whose error or omission could seriously harm a patient or user, including compromised care.

identify critical tasks

use error
An action or omission producing a result different from that intended or expected.

analyze a use error

close call
A situation in which a problem nearly occurs but is avoided.

document a close call

use difficulty
Observed trouble completing or understanding a task.

record a use difficulty

participant
A person taking part in a study.

recruit representative participants

moderator
The person guiding a study session under its protocol.

brief the moderator

prompting
Assistance that may influence a participant's performance.

document moderator prompting

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

human factors HUMAN FACTORS

apply human-factors principles

Keep the meaning precise: Study and application of how people interact with systems and products.

critical task CRITICAL TASK

identify critical tasks

Keep the meaning precise: A task whose error or omission could seriously harm a patient or user, including compromised care.

B Precision and professional shorthand

EXTEND YOUR RANGE

task success

Completion of a defined task under stated evaluation criteria.

define task success

root-cause analysis

Investigation of underlying contributors to an observed problem.

conduct root-cause analysis

affordance

A feature suggesting how an object or control can be used.

assess interface affordances

visual hierarchy

The arrangement that makes some information more prominent than other information.

improve the visual hierarchy

cognitive load

The mental effort needed to process information and perform a task.

reduce cognitive load

representative sample

A participant group suited to the intended study population and question.

justify the representative sample

observation note

A factual record of behavior or events during a study.

write an observation note

mental model

A user's understanding of how a system works.

explore the user's mental model

label comprehension

Understanding the meaning conveyed by a label.

evaluate label comprehension

legibility

How readily individual text or symbols can be distinguished.

assess label legibility

training decay

Loss of retained training knowledge or skill over time.

consider training decay

design iteration

A revised version created through successive development cycles.

evaluate a design iteration

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

prompting PROMPTING

document moderator prompting

Keep the meaning precise: Assistance that may influence a participant's performance.

affordance AFFORDANCE

assess interface affordances

Keep the meaning precise: A feature suggesting how an object or control can be used.

C Phrases that move the work forward

Report the observation

Three of eight participants selected the wrong status label.

Remove blame

The notes do not establish that the participants were careless.

Ask about wording

Did participants distinguish the meanings of the two labels?

Check assistance

Record whether the moderator prompted the participant.

Mark the revision

A larger font is proposed but has not been evaluated.

Compare explanations

Font size, wording, and hierarchy are different possible contributors.

Clarify task risk

Assess the task's safety significance through the appropriate process.

Request a scoped follow-up

Evaluate the revised label against the specific difficulty observed.

Preserve the context

This occurred in the simulated task used in the formative study.

Separate behavior from cause

The selection is observed; the reason still needs investigation.

Ask about visibility

Was the relevant text readily legible under the study conditions?

Bound the percentage

That percentage describes this study group, not the user population.

Avoid a success claim

We cannot yet say the change resolves the problem.

Keep evidence intact

Retain the observation notes when revising the summary.

Name the evaluation stage

These formative findings inform development; they are not a completed validation claim.

Close the report

State the finding, the open explanation, and the next evaluation separately.

Precision in real exchanges

Purpose: Separate behavior from cause
Useful wording: The selection is observed; the reason still needs investigation.

Purpose: Ask about wording
Useful wording: Did participants distinguish the meanings of the two labels?

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which summary is most accurate?

A. Three careless participants invalidate the study.

B. Larger text has already solved the issue.

C. Three of eight participants selected the wrong label in the formative simulated task.

D. The product has a proven 37.5% population error rate.

2. A participant can read both labels aloud but confuses their meanings. Which issue is most directly suggested?

A. Proven mechanical failure

B. Label comprehension

C. Inability to distinguish printed characters alone

D. Confirmed training decay in every user

3. A colleague needs the term for this meaning: An assessment of a sufficiently developed design against defined evaluation goals. Which term fits?

A. use difficulty

B. summative evaluation

C. cognitive load

D. root-cause analysis

D Choose the response that fits / continued

4. Rebuild the exchange from "Three participants read the label differently". Victor: "The team wants a larger font. That is easy to implement, but I cannot yet say that small characters caused all three wrong selections." Laila: [missing response] Victor: "One person read both labels aloud accurately and still selected the wrong status. Larger letters would not necessarily address that person's interpretation." Which response belongs in the gap?

A. Record the prompting. We need to see how completion was achieved; assisted success should not silently become unassisted task performance.

B. It is a visual hierarchy proposal. We can evaluate it alongside wording, but neither idea is a demonstrated solution yet.

C. Then keep the observation note factual: which label, which task, and what happened. Remove the character judgment without losing the actual difficulty.

D. Legibility is one question. We also need to examine whether the words distinguish the two states clearly in the task context.

Answers and explanations, page 131

> **One useful distinction**
>
> Three of eight is 37.5% of this study group, not an established error rate for the user population. Report the count and study context. A formative finding can identify a design concern without providing a final validation conclusion.

E Conversation

Three participants read the label differently

Three participants read the label differently

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

label comprehension / critical task / design iteration / Legibility / root-cause analysis / visual hierarchy / formative evaluation / human-factors validation / prompting / observation note

01 **Victor:** Three of the eight participants chose the wrong status label. The draft calls them careless. That is not something I can establish from watching the selection.

02 **Laila:** Then keep the [1] _____ factual: which label, which task, and what happened. Remove the character judgment without losing the actual difficulty.

03 **Victor:** Would thirty-seven point five percent sound more precise in the headline? I worry the count looks small, even though the issue may matter.

04 **Laila:** Identify the [2] _____ and keep three of eight visible. The percentage describes these participants; it is not an estimated population error rate.

05 **Victor:** The team wants a larger font. That is easy to implement, but I cannot yet say that small characters caused all three wrong selections.

06 **Laila:** [3] _____ is one question. We also need to examine whether the words distinguish the two states clearly in the task context.

07 **Victor:** One person read both labels aloud accurately and still selected the wrong status. Larger letters would not necessarily address that person's interpretation.

08 **Laila:** That observation concerns [4] _____. Keep it separate from a claim that the text could not be read, and do not assume every participant had the same problem.

09 **Victor:** Another person finished after the moderator asked a follow-up question. The results table currently says success, with no note about the intervention.

10 **Laila:** Record the [5] _____. We need to see how completion was achieved; assisted success should not silently become unassisted task performance.

Conversation continues on the next page.

Three participants read the label differently

CONVERSATION / CONTINUED

11 **Victor:** What about placing the status higher on the screen? That changes where attention goes, rather than only changing the text size.

12 **Laila:** It is a [6] _____ proposal. We can evaluate it alongside wording, but neither idea is a demonstrated solution yet.

13 **Victor:** I will stop calling the larger font a fix. The revised label has not been tested, and we may need a different approach.

14 **Laila:** Call it a proposed [7] _____. Link it to the observed difficulty and keep the follow-up result open rather than writing success in advance.

15 **Victor:** The project lead also asks whether a wrong selection makes this a critical task. Three errors alone do not answer the safety question, do they?

16 **Laila:** No. Designating a [8] _____ depends on potential consequences, including compromised care. The relevant risk assessment must address that, not just the number of selections.

17 **Victor:** Then I should not describe this as our final usability validation either. We used the session to find weaknesses while the interface was still developing.

18 **Laila:** Correct. Completed [9] _____ is not established by a formative session and an untested revision. State the evaluation stage and conditions clearly.

19 **Victor:** I will retain the three-of-eight count, the reading-versus-meaning observation, and the assisted completion. Proposed changes will sit in a separate section.

20 **Laila:** That gives the [10] _____ useful evidence. We can investigate the difficulty without blaming participants or promising that one easy design change resolves it.

CHECK THE GAPS

1. observation note / **2.** formative evaluation / **3.** Legibility / **4.** label comprehension / **5.** prompting / **6.** visual hierarchy / **7.** design iteration / **8.** critical task / **9.** human-factors validation / **10.** root-cause analysis

Why these answers fit, page 132

F Check accuracy and scope

Three participants read the label differently

Return to the case

Human-factors specialist Laila and interface designer Victor review a formative study of a fictional device label. Three of eight participants selected the wrong status label during a simulated task. The draft summary calls them careless and concludes that a larger font will solve the problem. No revised label has been tested, and the study does not establish a population error rate. Laila and Victor must preserve the observation, investigate the interface, and describe the proposed revision without declaring success.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. How should assistance during a task be reported?

A. Assume the moderator could not influence performance.

B. Document the prompting rather than presenting assisted success as unassisted performance.

C. Remove the assistance from the notes if the task was completed.

D. Treat any prompted completion as evidence that no difficulty existed.

2. Which next-status sentence fits the facts?

A. The revised label passed validation.

B. The study proves that only font size matters.

C. All intended users will now select correctly.

D. The larger-font revision is proposed; its effect on the observed difficulty remains untested.

Answers and explanations, page 134

Listen for the distinction

Swap roles. Contrast readable text with understood meaning and assisted completion with unassisted performance.

Describe the interaction before judging it

Careless assigns a personal explanation that an observation may not support. Describe what participants saw, did, and said, then examine the task and interface. Neutral language can be more demanding than blame because it requires the team to specify evidence for its explanation.

F Say it / transfer to a new case

Readable is not necessarily understood

Two of ten participants read a label accurately but misunderstand its meaning in a formative session. Revised wording is proposed and has not been tested.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

untested / session / resolved / meaning

01 **Researcher:** Two of ten had difficulty with the label's [1] _____.

02 **Designer:** This is an observation from a formative [2] _____.

03 **Researcher:** The new wording remains [3] _____.

04 **Designer:** We cannot yet claim the difficulty is [4] _____.

CHECK THE GAPS

1. meaning / **2.** session / **3.** untested / **4.** resolved

Why these answers fit, page 134

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Three participants read the label differently". Victor: "I will stop calling the larger font a fix. The revised label has not been tested, and we may need a different approach." Laila: [missing response] Victor: "The project lead also asks whether a wrong selection makes this a critical task. Three errors alone do not answer the safety question, do they?" Which response belongs in the gap?

A. No. Designating a critical task depends on potential consequences, including compromised care. The relevant risk assessment must address that, not just the number of selections.

B. Call it a proposed design iteration. Link it to the observed difficulty and keep the follow-up result open rather than writing success in advance.

C. Identify the formative evaluation and keep three of eight visible. The percentage describes these participants; it is not an estimated population error rate.

D. That gives the root-cause analysis useful evidence. We can investigate the difficulty without blaming participants or promising that one easy design change resolves it.

Answers and explanations, page 135

Review 2 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Verification, Validation, and Design Review" and "Usability and Human Factors". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A systematic evaluation of design progress, evidence, and issues. Which term fits?

A. simulated use

B. design review

C. requirement coverage

D. change impact

2. A colleague needs the term for this meaning: A situation in which a problem nearly occurs but is avoided. Which term fits?

A. visual hierarchy

B. close call

C. task success

D. human factors

3. Rebuild the exchange from "The hot surface is not the injury". Sam: "Should we remove the severity entry as well? It seems difficult to judge the risk while the likelihood information remains incomplete in this version." Elena: [missing response] Sam: "The cover remains attractive as an engineering proposal. I want to record it without giving the impression that it has already solved the issue." Which response belongs in the gap?

A. Keep severity separate from likelihood. We need to describe the possible injury carefully; uncertainty in one dimension does not make the other dimension disappear.

B. Not from this record. The accessible hot surface is the hazard. A skin burn describes a possible outcome, and the cover has not been evaluated.

C. We still need a justified probability of harm. An empty log does not establish zero occurrence, especially without understanding exposure, reporting, and the underlying evidence.

D. Yes, document the applicable risk acceptability criteria. The reviewer needs the reasoning and evidence, not simply a low label attached to an unfinished analysis.

Review 2 / Meaning and context / continued

4. Rebuild the exchange from "Three participants read the label differently". Victor: "Then I should not describe this as our final usability validation either. We used the session to find weaknesses while the interface was still developing." Laila: [missing response] Victor: "I will retain the three-of-eight count, the reading-versus-meaning observation, and the assisted completion. Proposed changes will sit in a separate section." Which response belongs in the gap?

A. Legibility is one question. We also need to examine whether the words distinguish the two states clearly in the task context.

B. Identify the formative evaluation and keep three of eight visible. The percentage describes these participants; it is not an estimated population error rate.

C. Correct. Completed human-factors validation is not established by a formative session and an untested revision. State the evaluation stage and conditions clearly.

D. Then keep the observation note factual: which label, which task, and what happened. Remove the character judgment without losing the actual difficulty.

Answers and explanations, page 159

A Read the situation

LESSON 05 / REGULATORY PATHWAYS AND SUBMISSIONS

Distinguish a proposed US submission pathway, supporting evidence, and an actual regulatory decision.

> **A route under review is not a clearance**
>
> Regulatory specialist Inez and commercial director Caleb review a fictional US launch plan. The team is evaluating a possible 510(k) pathway and has identified a candidate predicate, but its suitability is still under review. No submission has been filed and no FDA decision exists. A sales slide says FDA approved and promises an October launch. Inez must correct both the status and the certainty of the timeline while preserving a clear planning discussion.

People: Inez, Regulatory specialist; Caleb, Commercial director.

Read the record

Choose one answer per question using the facts above.

1. What is the current pathway status?

A. Every regulatory requirement has been waived.
B. A possible 510(k) route is being evaluated.
C. FDA has granted a PMA approval.
D. A 510(k) clearance has been issued.

2. What is known about the predicate?

A. Its suitability has been conclusively accepted by FDA.
B. It automatically covers every proposed use.
C. No comparison is needed because the devices look similar.
D. A candidate has been identified, but suitability remains under review.

3. Why is the October launch promise unsupported?

A. October launches are never possible.
B. No filing or regulatory decision exists, and the pathway assessment is incomplete.
C. All 510(k) reviews are legally guaranteed to fail.
D. The product has already been prohibited permanently.

Answers and explanations, page 136

B Vocabulary in use

CORE TERMS

regulatory pathway
The applicable route for obtaining a required regulatory decision.

assess the regulatory pathway

device classification
A regulatory category reflecting the device type and applicable controls.

confirm device classification

product code
A code used by FDA to identify a device type.

verify the product code

510(k)
A US premarket notification seeking a substantial-equivalence determination.

prepare a 510(k)

predicate device
A legally marketed comparator used in a 510(k) assessment.

assess a candidate predicate

substantial equivalence
The comparative regulatory determination central to a 510(k) decision.

support substantial equivalence

De Novo request
A US request for risk-based classification of a novel eligible device.

assess a De Novo request

premarket approval
The US PMA pathway requiring an independent safety-and-effectiveness demonstration.

prepare a premarket approval application

clearance
A favorable 510(k) determination permitting marketing within its scope.

obtain 510(k) clearance

approval
A formal favorable decision under an applicable approval pathway.

distinguish approval from clearance

indications for use
The specific purposes and population described for a device's use.

define indications for use

technological characteristics
Design and operating features relevant to a regulatory comparison.

compare technological characteristics

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

regulatory pathway REGULATORY PATHWAY

assess the regulatory pathway

Keep the meaning precise: The applicable route for obtaining a required regulatory decision.

substantial equivalence SUBSTANTIAL EQUIVALENCE

support substantial equivalence

Keep the meaning precise: The comparative regulatory determination central to a 510(k) decision.

B Precision and professional shorthand

EXTEND YOUR RANGE

general controls

Baseline regulatory controls applicable to medical devices.

comply with general controls

special controls

Additional controls applicable to certain device types.

identify special controls

submission dossier

The organized documents and evidence submitted for regulatory review.

assemble a submission dossier

evidence plan

A plan for generating and organizing support for specified claims.

develop an evidence plan

regulatory strategy

A coordinated approach to regulatory requirements and decisions.

align the regulatory strategy

pre-submission interaction

A planned exchange with FDA before a later submission.

request a pre-submission interaction

review question

A request for clarification or evidence during review.

address a review question

deficiency

A shortcoming identified in the submitted information.

respond to a deficiency

response package

Documents addressing specified reviewer questions or deficiencies.

compile a response package

marketing authorization

The applicable permission to market a device within a defined scope.

confirm marketing authorization

regulatory dependency

An unresolved regulatory matter affecting another planned step.

identify a regulatory dependency

launch assumption

A planning premise underlying the proposed market-entry date.

qualify a launch assumption

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

technological characteristics TECHNOLOGICAL CHARACTERISTICS

compare technological characteristics

Keep the meaning precise: Design and operating features relevant to a regulatory comparison.

regulatory strategy REGULATORY STRATEGY

align the regulatory strategy

Keep the meaning precise: A coordinated approach to regulatory requirements and decisions.

C Phrases that move the work forward

State the current stage
We are evaluating a possible 510(k) pathway.

Correct the decision claim
No FDA decision has been issued for this device.

Use the right term
A favorable 510(k) determination is clearance, not PMA approval.

Clarify the claim
The proposed indications must be consistent across the evidence and materials.

Identify dependencies
The pathway assessment and evidence requirements remain unresolved.

Request specialist review
Regulatory needs to assess the proposed use and comparison.

Align the slide
Replace FDA approved with the actual development status.

Name the next deliverable
The next step is the pathway assessment, not a launch announcement.

Qualify the comparator
The predicate candidate remains under review.

Name the missing filing
The submission has not yet been filed.

Separate similarity from evidence
Similar appearance alone does not establish substantial equivalence.

Bound the timeline
October is a planning assumption, not an authorized launch date.

Avoid predicting rejection
An open assessment does not mean the device has been rejected.

Describe an interaction
A pre-submission discussion is not a marketing authorization.

Keep decisions traceable
Use the exact decision and scope when an authorization exists.

Close the planning discussion
We can plan around dependencies without promising the agency's decision.

Precision in real exchanges

Purpose: Name the missing filing
Useful wording: The submission has not yet been filed.

Purpose: Use the right term
Useful wording: A favorable 510(k) determination is clearance, not PMA approval.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which sales-slide correction matches the case?

A. PMA rejected because the team discussed 510(k).

B. Possible 510(k) pathway under internal assessment; no submission or FDA decision yet.

C. FDA approved because a predicate was found.

D. FDA clearance guaranteed by October.

2. Which statement about the candidate predicate is accurate?

A. Any similar-looking marketed product guarantees substantial equivalence.

B. A candidate predicate grants marketing permission.

C. A product code alone proves every proposed indication is supported.

D. Identification begins the assessment; suitability and the comparative evidence still need review.

3. A colleague needs the term for this meaning: A US premarket notification seeking a substantial-equivalence determination. Which term fits?

A. response package

B. 510(k)

C. clearance

D. submission dossier

D Choose the response that fits / continued

4. Rebuild the exchange from "A route under review is not a clearance". Caleb: "What should the commercial team understand about that comparison? I want the next discussion to focus on the evidence rather than repeat the same assumption." Inez: [missing response] Caleb: "Nothing has been filed, correct? Then under review is too vague; sales could read that as an FDA review already underway." Which response belongs in the gap?

A. Yes, keep the regulatory strategy provisional where appropriate. This discussion does not determine eligibility for another route, and we should not guess the outcome from limited facts.

B. The relevant determination is substantial equivalence. The proposed use and technological characteristics matter, and the evidence must support the actual comparison rather than a convenient headline.

C. Neither is an authorization. We are still assessing the regulatory pathway, including whether a 510(k) route is appropriate for the proposed device and its claims.

D. It is a candidate predicate device. Its suitability remains under review; visual similarity alone does not establish the regulatory comparison needed for the proposed use.

Answers and explanations, page 136

> **One useful distinction**
>
> For a US 510(k), clearance is the relevant decision term; it should not be casually called PMA approval. A candidate predicate, a pathway assessment, a submitted file, and an agency decision are four different statuses.

E Conversation

A route under review is not a clearance

A route under review is not a clearance

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

regulatory pathway / clearance / pre-submission interaction / substantial equivalence / submission dossier / predicate device / regulatory strategy / indications for use / marketing authorization / launch assumption

01 **Caleb:** The launch slide says FDA approved and shows October as the launch month. I need to understand whether those statements reflect an actual decision or our working plan.

02 **Inez:** Neither is an authorization. We are still assessing the [1] _____, including whether a 510(k) route is appropriate for the proposed device and its claims.

03 **Caleb:** The team found a marketed device that looks similar. I took that as confirmation that the route was settled, but the comparison may not be that simple.

04 **Inez:** It is a candidate [2] _____. Its suitability remains under review; visual similarity alone does not establish the regulatory comparison needed for the proposed use.

05 **Caleb:** What should the commercial team understand about that comparison? I want the next discussion to focus on the evidence rather than repeat the same assumption.

06 **Inez:** The relevant determination is [3] _____. The proposed use and technological characteristics matter, and the evidence must support the actual comparison rather than a convenient headline.

E Conversation / continued

07 **Caleb:** Nothing has been filed, correct? Then under review is too vague; sales could read that as an FDA review already underway.

08 **Inez:** Exactly. The [4] _____ is not filed. Say internal pathway assessment, and keep that separate from any future agency review or decision.

09 **Caleb:** If the team eventually follows the 510(k) route successfully, should the slide say approved? That term is common in sales conversations, but I want it to be accurate.

10 **Inez:** Use [5] _____ for the actual favorable 510(k) determination. Do not borrow the language of a different pathway or imply a decision before one exists.

Conversation continues on the next page.

A route under review is not a clearance

CONVERSATION / CONTINUED

11 **Caleb:** Could a different route still be needed? I do not want the revised slide to make the 510(k) plan sound irrevocable while the assessment is incomplete.

12 **Inez:** Yes, keep the [6] _____ provisional where appropriate. This discussion does not determine eligibility for another route, and we should not guess the outcome from limited facts.

13 **Caleb:** The product team also wants broader wording about who can use the device. That seems connected to the pathway work, not just to marketing style.

14 **Inez:** It is. The [7] _____ must be clear and consistent. A broader statement can change the claim being assessed and the evidence needed to support it.

15 **Caleb:** Then October should be shown as an internal target with conditions. We cannot treat it as an agency commitment when neither the pathway nor the evidence package is settled.

16 **Inez:** Call it a [8] _____ and show the dependencies. That allows planning without making a promise about a regulatory decision that has not occurred.

17 **Caleb:** We may seek feedback before filing. I will make sure the team does not describe a discussion with the agency as permission to market the device.

18 **Inez:** Correct. A [9] _____ is not the marketing decision. Any feedback and its scope should be recorded accurately without converting it into blanket approval.

19 **Caleb:** I will remove FDA approved and mark October as conditional. Send me the pathway-assessment milestone and its dependencies for the planning slide.

20 **Inez:** Good. The communication must distinguish planning from [10] _____. We can describe the work confidently while leaving the unresolved regulatory decision exactly where it belongs.

CHECK THE GAPS

1. regulatory pathway / **2.** predicate device / **3.** substantial equivalence / **4.** submission dossier / **5.** clearance / **6.** regulatory strategy / **7.** indications for use / **8.** launch assumption / **9.** pre-submission interaction / **10.** marketing authorization

Why these answers fit, page 137

F Check accuracy and scope

A route under review is not a clearance

Return to the case

Regulatory specialist Inez and commercial director Caleb review a fictional US launch plan. The team is evaluating a possible 510(k) pathway and has identified a candidate predicate, but its suitability is still under review. No submission has been filed and no FDA decision exists. A sales slide says FDA approved and promises an October launch. Inez must correct both the status and the certainty of the timeline while preserving a clear planning discussion.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which October-launch note preserves the current regulatory status?

A. Confirmed market-entry date once the candidate predicate is listed in the slide

B. Agency review deadline beginning when the internal pathway discussion started

C. Internal planning target; pathway and predicate still under assessment, with no filing or decision

D. Expected FDA clearance date because the commercial team has approved the target

2. What does a pre-submission interaction establish by itself?

A. A guaranteed review outcome

B. Permission for every proposed claim

C. An opportunity for feedback, not marketing authorization

D. A final 510(k) clearance

Answers and explanations, page 139

Listen for the distinction

Swap roles. Use clearance for the 510(k) outcome and describe October as an internal conditional target.

Correct status without inventing the opposite

An unsupported approval claim should be replaced with the actual status, not with a prediction that the device will fail. Separate the pathway assessment, evidence plan, submission, review, and decision. This gives commercial colleagues information they can use without a guarantee the team cannot make.

F Say it / transfer to a new case

A filing is not a favorable decision

A fictional US 510(k) submission has been filed. Review is ongoing, and no decision has been issued. The commercial team has an internal target date.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

filed / clearance / ongoing / assumption

01 **Lead:** The submission has been [1] _____.

02 **Specialist:** The agency's review remains [2] _____.

03 **Lead:** No 510(k) [3] _____ has been issued.

04 **Specialist:** The target date is a planning [4] _____.

CHECK THE GAPS

1. filed / **2.** ongoing / **3.** clearance / **4.** assumption

Why these answers fit, page 139

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A route under review is not a clearance". Caleb: "The product team also wants broader wording about who can use the device. That seems connected to the pathway work, not just to marketing style." Inez: [missing response] Caleb: "Then October should be shown as an internal target with conditions. We cannot treat it as an agency commitment when neither the pathway nor the evidence package is settled." Which response belongs in the gap?

A. Yes, keep the regulatory strategy provisional where appropriate. This discussion does not determine eligibility for another route, and we should not guess the outcome from limited facts.

B. It is. The indications for use must be clear and consistent. A broader statement can change the claim being assessed and the evidence needed to support it.

C. Neither is an authorization. We are still assessing the regulatory pathway, including whether a 510(k) route is appropriate for the proposed device and its claims.

D. The relevant determination is substantial equivalence. The proposed use and technological characteristics matter, and the evidence must support the actual comparison rather than a convenient headline.

Answers and explanations, page 140

A Read the situation

LESSON 06 / COMPLAINTS, MDRS, AND POSTMARKET SIGNALS

Capture a complaint accurately, route missing information promptly, and distinguish counts from rates and reports from causation.

> **Incomplete information still needs a handoff**
>
> Complaint specialist Nora and service lead Dev review a caller's report that a fictional monitor unexpectedly stopped displaying. The event date, device identifier, and patient outcome are not yet known. The call was received today at 14:10. Dev proposes waiting for every field before routing it. A separate dashboard shows eight complaints this quarter versus four last quarter, without use-volume data. Nora must preserve the unknowns, arrange the required internal review, and prevent either dataset from implying an unsupported conclusion.

People: Nora, Complaint specialist; Dev, Service lead.

Read the record

Choose one answer per question using the facts above.

1. Which time is known?

A. The time FDA issued a decision
B. The call-receipt time, 14:10 today
C. The exact event time
D. The time of a confirmed patient injury

2. What is known about patient outcome?

A. No injury occurred.
B. A serious injury is confirmed.
C. The device definitely caused harm.
D. It is not yet known.

3. What does eight versus four establish?

A. The product caused twice as many injuries.
B. Reporting behavior stayed identical.
C. The complaint count doubled, but no rate can be calculated from the supplied data.
D. The rate per use doubled.

Answers and explanations, page 141

B Vocabulary in use

CORE TERMS

complaint
A reported concern about a device's quality, performance, safety, or related service.

capture a complaint

adverse event
An unfavorable occurrence that may require assessment of its relationship to a device.

assess an adverse event

malfunction
A failure to perform as intended or meet relevant performance specifications.

investigate a reported malfunction

Medical Device Reporting
The US system for reporting specified device events under applicable requirements.

assess Medical Device Reporting obligations

reportability
Whether an event meets applicable reporting requirements.

assess reportability

initial reporter
The person or organization first providing event information.

contact the initial reporter

awareness date
The date relevant event information becomes known for the applicable assessment.

document the awareness date

event date
The date on which the reported occurrence took place.

confirm the event date

receipt date
The date the organization received the communication.

record the receipt date

Unique Device Identifier
A device-identification code with defined identification components.

capture the Unique Device Identifier

serial number
An identifier distinguishing an individual unit.

verify the serial number

lot number
An identifier linking items to a production batch.

record the lot number

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

complaint COMPLAINT

capture a complaint

Keep the meaning precise: A reported concern about a device's quality, performance, safety, or related service.

initial reporter INITIAL REPORTER

contact the initial reporter

Keep the meaning precise: The person or organization first providing event information.

B Precision and professional shorthand

EXTEND YOUR RANGE

patient outcome

The reported effect on the patient's condition or health.

clarify the patient outcome

event narrative

A factual account of the reported occurrence and its context.

preserve the event narrative

follow-up attempt

A documented effort to obtain missing or clarifying information.

record a follow-up attempt

postmarket surveillance

Systematic collection and review of information after market introduction.

conduct postmarket surveillance

trend analysis

Review of patterns in data over time.

perform trend analysis

duplicate report

Another record potentially describing an already reported event.

assess duplicate reports

causal relationship

A supported connection between a device and an observed event.

assess the causal relationship

reasonably known information

Information available or obtainable as defined by the applicable reporting requirement.

provide reasonably known information

supplemental report

A later report providing required additional information.

submit a supplemental report

safety signal

Information suggesting a potential safety issue that needs assessment.

investigate a safety signal

exposure denominator

The relevant use or exposure quantity against which events are compared.

identify the exposure denominator

complaint closure

Documented completion of the required complaint-review process.

authorize complaint closure

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

lot number LOT NUMBER

record the lot number

Keep the meaning precise: An identifier linking items to a production batch.

follow-up attempt FOLLOW-UP ATTEMPT

record a follow-up attempt

Keep the meaning precise: A documented effort to obtain missing or clarifying information.

C Phrases that move the work forward

Capture the account

The caller reported that the display stopped unexpectedly.

Mark a missing identifier

We have not yet confirmed the device identifier.

Avoid causal certainty

This report does not by itself establish the cause.

Continue follow-up

Record the questions asked and the attempts to obtain missing information.

Avoid a blanket rule

Missing fields do not justify withholding the initial internal handoff.

State the count

The count increased from four to eight.

Avoid dismissal

An incomplete account still needs appropriate review.

Name the update route

New required information must follow the applicable reporting process.

Separate dates

The call was received today; the event date is still unknown.

Preserve outcome uncertainty

Patient outcome has not been established.

Route available facts

Send the available information for prompt internal assessment.

Distinguish the review

Reportability needs assessment under the applicable requirements.

Use the right denominator

We need the relevant exposure data before describing a rate.

Check duplicates

Confirm whether separate records describe the same event.

Preserve the original

Keep the initial narrative alongside later clarifications.

Close with status

The case is routed, follow-up is open, and no cause is confirmed.

Precision in real exchanges

Purpose: Preserve outcome uncertainty
Useful wording: Patient outcome has not been established.

Purpose: Avoid causal certainty
Useful wording: This report does not by itself establish the cause.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which intake sentence preserves the facts?

A. Case closed because the serial number is missing.

B. Display interruption reported; event date, identifier, and patient outcome unknown; call received today at 14:10.

C. No injury and no malfunction; event occurred today at 14:10.

D. Confirmed device-caused injury at an unknown hospital.

2. What is the appropriate internal handoff wording?

A. Reject the report because the caller lacks an identifier.

B. Declare it nonreportable without the required assessment.

C. Route the available account promptly for assessment and continue documented follow-up.

D. Hold the case indefinitely until every field is complete.

3. A colleague needs the term for this meaning: The US system for reporting specified device events under applicable requirements. Which term fits?

A. trend analysis

B. receipt date

C. Medical Device Reporting

D. event narrative

D Choose the response that fits / continued

4. Rebuild the exchange from "Incomplete information still needs a handoff". Dev: "The caller did not tell us whether anyone was injured. The draft summary says no injury, which is more definite than the information actually supports." Nora: [missing response] Dev: "We also need to identify the unit accurately. A model name alone might not let the investigation team distinguish this device from other units at the site." Which response belongs in the gap?

A. We should route the available event narrative for prompt internal assessment and continue follow-up. An incomplete account is not a reason to withhold it from the responsible team.

B. Yes, assess each possible duplicate report while preserving the source information. Deduplication needs a basis; a similar description alone should not make us erase a separate event.

C. Then the causal relationship remains unconfirmed. Report the observation accurately without either asserting device causation or dismissing the device as unrelated before the evidence supports that conclusion.

D. Change patient outcome to unknown. We should ask for clarification and record the response without treating silence in the first call as a confirmed absence of harm.

Answers and explanations, page 141

One useful distinction

The receipt time is not the event time. Unknown patient outcome is not the same as no injury. Preserve the reporter's account and route the available information through the applicable process while actively seeking missing facts.

E Conversation

Incomplete information still needs a handoff

Incomplete information still needs a handoff

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

patient outcome / supplemental report / Unique Device Identifier / causal relationship / duplicate report / exposure denominator / reportability / receipt date / follow-up attempt / event narrative

01 **Dev:** The caller said the monitor stopped displaying, but the device identifier and event date are missing. I thought we should wait before sending an incomplete record onward.

02 **Nora:** We should route the available [1] _____ for prompt internal assessment and continue follow-up. An incomplete account is not a reason to withhold it from the responsible team.

03 **Dev:** I have routed the account now. The call arrived today at fourteen ten; the event date is still unknown.

04 **Nora:** I have received it for assessment. Record the [2] _____ and time separately; do not copy them into the unknown event-date field.

05 **Dev:** The caller did not tell us whether anyone was injured. The draft summary says no injury, which is more definite than the information actually supports.

06 **Nora:** Change [3] _____ to unknown. We should ask for clarification and record the response without treating silence in the first call as a confirmed absence of harm.

E Conversation / continued

07 **Dev:** We also need to identify the unit accurately. A model name alone might not let the investigation team distinguish this device from other units at the site.

08 **Nora:** Seek the [4] _____ and other relevant identification details through the approved process. Record what is available rather than inventing a serial number to complete the form.

09 **Dev:** Is this automatically reportable because the display stopped? Service needs to avoid delay without deciding beyond its role.

10 **Nora:** The responsible team must assess [5] _____ under the applicable requirements. Preserve the account and route it promptly; do not replace the assessment with an automatic yes or no.

Conversation continues on the next page.

Incomplete information still needs a handoff

CONVERSATION / CONTINUED

11 **Dev:** I will document the missing questions and the calls we make to obtain answers. The case should show what we tried, not simply remain blank.

12 **Nora:** Each [6] _____ helps make the information history clear. New facts should be linked to their source, including when they became available and any earlier uncertainty they resolve.

13 **Dev:** The summary says device-caused. Where did that conclusion come from? The caller described the display stopping; we do not yet have the cause.

14 **Nora:** Then the [7] _____ remains unconfirmed. Report the observation accurately without either asserting device causation or dismissing the device as unrelated before the evidence supports that conclusion.

15 **Dev:** The quarterly dashboard adds another concern: eight complaints this quarter and four last quarter. The presentation says the rate doubled, but there is no use-volume information.

16 **Nora:** We lack the [8] _____. The count doubled; the rate per use is not established. We also need to examine reporting patterns and whether the records are comparable.

17 **Dev:** Some records might refer to the same occurrence. We should check that before treating every entry as a separate event or comparing the clinical outcomes.

18 **Nora:** Yes, assess each possible [9] _____ while preserving the source information. Deduplication needs a basis; a similar description alone should not make us erase a separate event.

Incomplete information still needs a handoff / continued

19 **Dev:** The account is routed and you have confirmed receipt. I will correct the unknown fields and retain eight versus four complaints without inventing a rate or injury outcome.

20 **Nora:** Good. Any required later information must follow the applicable
10 _____ process. Keep investigation, reporting assessment, and complaint closure distinct so the record never suggests an unfinished task has already been completed.

CHECK THE GAPS

1. event narrative / **2.** receipt date / **3.** patient outcome / **4.** Unique Device Identifier / **5.** reportability / **6.** follow-up attempt / **7.** causal relationship / **8.** exposure denominator / **9.** duplicate report / **10.** supplemental report

Why these answers fit, page 142

F Check accuracy and scope

Incomplete information still needs a handoff

Return to the case

Complaint specialist Nora and service lead Dev review a caller's report that a fictional monitor unexpectedly stopped displaying. The event date, device identifier, and patient outcome are not yet known. The call was received today at 14:10. Dev proposes waiting for every field before routing it. A separate dashboard shows eight complaints this quarter versus four last quarter, without use-volume data. Nora must preserve the unknowns, arrange the required internal review, and prevent either dataset from implying an unsupported conclusion.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. What does the dashboard support?

A. The injury rate doubled exactly.

B. Device causation doubled exactly.

C. Eight complaints prove eight unique serious injuries.

D. The count rose from four to eight; the rate per use is not established.

2. The caller later reports no injury. Which update preserves the information history and review responsibilities?

A. Replace the original unknown field as though no injury was established at first receipt.

B. Close reporting assessment automatically because no injury is now reported.

C. Remove the original narrative because a later account supersedes every earlier detail.

D. Add the attributed clarification and its receipt time; continue the applicable reporting assessment.

Answers and explanations, page 144

Listen for the distinction

Swap roles. Say outcome unknown rather than no injury, and doubled count rather than doubled use-based rate.

Unknown is a useful status

A missing field can be stated clearly without sounding evasive. Say what was reported, what is unknown, who is seeking it, and where the information has been routed. This supports timely review while avoiding a false conclusion that no injury or device problem occurred.

F Say it / transfer to a new case

The report date is known

A caller reports a device issue today. The event date and outcome are unknown. Six complaints were recorded this month, but the number of uses is unavailable.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

received / count / denominator / unknown

01 **Specialist:** Today is the date the report was [1] _____.

02 **Colleague:** The patient outcome is still [2] _____.

03 **Specialist:** Six is the complaint [3] _____.

04 **Colleague:** A use-based rate needs the exposure [4] _____.

CHECK THE GAPS

1. received / **2.** unknown / **3.** count / **4.** denominator

Why these answers fit, page 144

F Say it / transfer to a new case / continued

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Incomplete information still needs a handoff". Dev: "The summary says device-caused. Where did that conclusion come from? The caller described the display stopping; we do not yet have the cause." Nora: [missing response] Dev: "The quarterly dashboard adds another concern: eight complaints this quarter and four last quarter. The presentation says the rate doubled, but there is no use-volume information." Which response belongs in the gap?

A. Then the causal relationship remains unconfirmed. Report the observation accurately without either asserting device causation or dismissing the device as unrelated before the evidence supports that conclusion.

B. Change patient outcome to unknown. We should ask for clarification and record the response without treating silence in the first call as a confirmed absence of harm.

C. We should route the available event narrative for prompt internal assessment and continue follow-up. An incomplete account is not a reason to withhold it from the responsible team.

D. Yes, assess each possible duplicate report while preserving the source information. Deduplication needs a basis; a similar description alone should not make us erase a separate event.

Answers and explanations, page 145

Review 3 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Regulatory Pathways and Submissions" and "Complaints, MDRs, and Postmarket Signals". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A code used by FDA to identify a device type. Which term fits?

A. deficiency
B. premarket approval
C. special controls
D. product code

2. A colleague needs the term for this meaning: The date on which the reported occurrence took place. Which term fits?

A. postmarket surveillance
B. event date
C. complaint
D. patient outcome

3. Rebuild the exchange from "Three participants read the label differently". Victor: "Another person finished after the moderator asked a follow-up question. The results table currently says success, with no note about the intervention." Laila: [missing response] Victor: "What about placing the status higher on the screen? That changes where attention goes, rather than only changing the text size." Which response belongs in the gap?

A. Legibility is one question. We also need to examine whether the words distinguish the two states clearly in the task context.
B. Then keep the observation note factual: which label, which task, and what happened. Remove the character judgment without losing the actual difficulty.
C. Record the prompting. We need to see how completion was achieved; assisted success should not silently become unassisted task performance.
D. It is a visual hierarchy proposal. We can evaluate it alongside wording, but neither idea is a demonstrated solution yet.

Review 3 / Meaning and context / continued

4. Rebuild the exchange from "Incomplete information still needs a handoff". Dev: "Some records might refer to the same occurrence. We should check that before treating every entry as a separate event or comparing the clinical outcomes." Nora: [missing response] Dev: "The account is routed and you have confirmed receipt. I will correct the unknown fields and retain eight versus four complaints without inventing a rate or injury outcome." Which response belongs in the gap?

A. Change patient outcome to unknown. We should ask for clarification and record the response without treating silence in the first call as a confirmed absence of harm.

B. We should route the available event narrative for prompt internal assessment and continue follow-up. An incomplete account is not a reason to withhold it from the responsible team.

C. Yes, assess each possible duplicate report while preserving the source information. Deduplication needs a basis; a similar description alone should not make us erase a separate event.

D. Then the causal relationship remains unconfirmed. Report the observation accurately without either asserting device causation or dismissing the device as unrelated before the evidence supports that conclusion.

Answers and explanations, page 162

A Read the situation

LESSON 07 / MANUFACTURING, SUPPLIERS, AND NONCONFORMANCE

Separate affected-lot scope, observed failures, containment, cause, and authorized disposition in a supplier discussion.

Three hundred pieces on hold

Quality engineer Tomas and supply planner Grace review two incoming component lots containing 120 and 180 pieces. One sampled piece from each lot failed a specified dimensional check. Both lots are on hold under the fictional site procedure. The cause is not established, and no release or rework disposition is authorized. Production needs parts tomorrow. Grace proposes describing all 300 pieces as defective, then releasing the second lot because its supplier certificate says conforming.

People: Tomas, Quality engineer; Grace, Supply planner.

Read the record

Choose one answer per question using the facts above.

1. How many pieces are in the two held lots?

A. 2
B. 120
C. 180
D. 300

2. How many sampled pieces are explicitly reported as failing?

A. Every piece in the second lot
B. None because a certificate exists
C. Two, one from each lot
D. All 300

3. What is the current disposition?

A. The second lot is released automatically.
B. Both lots must be reworked using any available method.
C. All pieces are confirmed scrap.
D. Both lots remain on hold; no release or rework is authorized.

Answers and explanations, page 146

B Vocabulary in use

CORE TERMS

incoming inspection
Examination of received material against specified requirements.

perform incoming inspection

nonconforming material
Material identified as not meeting a specified requirement.

control nonconforming material

quarantine
Controlled segregation or status preventing unintended use.

place material in quarantine

rework
Action intended to bring nonconforming product into conformity.

authorize a rework instruction

concession
An authorized permission to use or release specified nonconforming product under defined conditions.

evaluate a concession request

supplier corrective action request
A formal request for a supplier to address a quality problem.

issue a supplier corrective action request

nonconformance
Failure to fulfill a specified requirement.

document a nonconformance

containment
Immediate action to limit the scope or spread of a problem.

implement containment

disposition
An authorized decision on how identified material will be handled.

approve material disposition

scrap
Material designated not to be used for its intended production purpose.

document a scrap decision

certificate of conformity
A supplier statement that supplied material meets specified requirements.

review the certificate of conformity

corrective action
Action addressing the cause of a nonconformity to prevent recurrence.

verify corrective action

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

incoming inspection INCOMING INSPECTION

perform incoming inspection

Keep the meaning precise: Examination of received material against specified requirements.

disposition DISPOSITION

approve material disposition

Keep the meaning precise: An authorized decision on how identified material will be handled.

B Precision and professional shorthand

EXTEND YOUR RANGE

correction

Action addressing a detected nonconformity itself.

distinguish correction from corrective action

lot genealogy

Records linking material lots through production and distribution.

trace lot genealogy

dimensional tolerance

The permitted variation in a specified measurement.

check dimensional tolerance

calibration status

Whether a measuring instrument's calibration requirements are satisfied.

verify calibration status

change notification

Communication of a proposed or implemented change affecting supply.

require a change notification

CAPA

Corrective and preventive action within a quality-system process.

assess the need for CAPA

acceptance sampling

Evaluation of a lot using a defined sample and decision plan.

apply an acceptance-sampling plan

measurement uncertainty

The quantified uncertainty associated with a measurement result.

evaluate measurement uncertainty

supplier qualification

Evaluation and approval of a supplier for a defined purpose.

maintain supplier qualification

release status

The current authorization state for using or distributing material.

verify release status

B Precision and professional shorthand / continued

Quality Management System Regulation

The current US device quality-system regulation, commonly called QMSR.

follow the Quality Management System Regulation

ISO 13485

An international standard for medical-device quality management systems.

apply ISO 13485 requirements

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

corrective action CORRECTIVE ACTION

verify corrective action

Keep the meaning precise: Action addressing the cause of a nonconformity to prevent recurrence.

dimensional tolerance DIMENSIONAL TOLERANCE

check dimensional tolerance

Keep the meaning precise: The permitted variation in a specified measurement.

C Phrases that move the work forward

State the held population

Both lots are on hold, totaling 300 pieces.

State the observed failures

One sampled piece from each lot failed the dimensional check.

Avoid overstating scope

We have not inspected and confirmed every piece as defective.

Avoid assuming conformity

The uninspected pieces are not thereby proven conforming.

Name the requirement

Reference the specified dimension and applicable tolerance.

Preserve the hold

No release or rework disposition is authorized.

Acknowledge urgency

I understand tomorrow's need; the material status still has to be respected.

Assess contrary evidence

The certificate does not erase the failed inspection result.

Separate containment

The hold limits unintended use while the issue is assessed.

Keep cause open

The supplier cause has not been established.

Request a response

Ask for evidence addressing the specified nonconformance.

Clarify corrective action

Fixing a piece and preventing recurrence are different objectives.

Verify the instrument

Include calibration status and relevant measurement information in the review.

Trace the scope

Use lot records to assess what other material may be affected.

Define authorization

Only the designated process can approve the material disposition.

Close the handoff

Give production the confirmed status and the next review time, not an assumed release.

Precision in real exchanges

Purpose: Avoid assuming conformity
Useful wording: The uninspected pieces are not thereby proven conforming.

Purpose: Name the requirement
Useful wording: Reference the specified dimension and applicable tolerance.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which quantity statement is accurate?

A. Only two pieces are held.

B. The 180-piece lot has no failed result.

C. Three hundred pieces are held; two sampled pieces are explicitly reported as failing.

D. All 300 pieces were individually tested and failed.

2. Production proposes using the 180-piece lot because its certificate is conforming. Which response preserves the current evidence and authority?

A. The production deadline supplies temporary release authority pending the investigation.

B. The failed sample and hold remain; the certificate does not replace an authorized disposition.

C. The certificate releases the lot, while only its failed sample remains on hold.

D. The lot is usable because the other 179 pieces have not been reported as failed.

3. A colleague needs the term for this meaning: Immediate action to limit the scope or spread of a problem. Which term fits?

A. lot genealogy

B. concession

C. containment

D. change notification

D Choose the response that fits / continued

4. Rebuild the exchange from "Three hundred pieces on hold". Grace: "The 180-piece lot has a supplier certificate saying conforming. Can that support using it while we investigate the failed sample?" Tomas: [missing response] Grace: "I will tell production both lots remain unavailable. We have not established whether the two sampled failures have the same cause." Which response belongs in the gap?

A. Correct. Link each nonconformance to the dimensional requirement and lot. Uninspected does not mean conforming, either; keep the hold status visible.

B. Incoming inspection found two failed sampled pieces, one per lot. Both lots are held; those are different quantities and different statements.

C. Exactly. containment can be in place while cause remains open. Controlling the material does not require us to pretend the investigation is complete.

D. The certificate of conformity does not override contrary inspection evidence. That lot remains held unless the authorized process establishes a different disposition.

Answers and explanations, page 146

> **One useful distinction**
>
> The held population is 300 pieces. The supplied observation is two failed sampled pieces, one from each lot. Do not call all 300 individually confirmed defective, or treat uninspected pieces as conforming. The hold and the observed failures describe different scopes.

E Conversation

Three hundred pieces on hold

Three hundred pieces on hold

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

release status / correction / containment / disposition / lot genealogy / certificate of conformity / corrective action / calibration status / Incoming inspection / nonconformance

01 **Grace:** Production needs the parts tomorrow. The lots contain 120 and 180 pieces. I was about to report all 300 as defective, but that overstates our inspection.

02 **Tomas:** [1] _____ found two failed sampled pieces, one per lot. Both lots are held; those are different quantities and different statements.

03 **Grace:** So the report needs three hundred held, two observed failures. We cannot assign an individual result to pieces that were not inspected.

04 **Tomas:** Correct. Link each [2] _____ to the dimensional requirement and lot. Uninspected does not mean conforming, either; keep the hold status visible.

05 **Grace:** The 180-piece lot has a supplier certificate saying conforming. Can that support using it while we investigate the failed sample?

06 **Tomas:** The [3] _____ does not override contrary inspection evidence. That lot remains held unless the authorized process establishes a different disposition.

07 **Grace:** I will tell production both lots remain unavailable. We have not established whether the two sampled failures have the same cause.

08 **Tomas:** Exactly. [4] _____ can be in place while cause remains open. Controlling the material does not require us to pretend the investigation is complete.

09 **Grace:** Could I ask quality about rework tonight? I need to offer production a next step without accidentally authorizing work myself.

10 **Tomas:** Request a [5] _____ review. Do not tell anyone to rework or use material before the required decision, instructions, and evaluation are in place.

Conversation continues on the next page.

Three hundred pieces on hold

CONVERSATION / CONTINUED

11 **Grace:** Then tonight is a request for review, not an expected release. I should give production the actual status and the next update we can commit to.

12 **Tomas:** Keep [6] _____ separate from the deadline. Schedule pressure can make the review urgent; it cannot provide evidence or approval.

13 **Grace:** The supplier offers to replace the two failed samples. That might address those particular pieces, but does not explain the dimensional problem.

14 **Tomas:** Replacement could be a [7] _____ through the approved process. It would not by itself establish the cause or prevent the same issue recurring.

15 **Grace:** I will request their manufacturing and inspection records. We should also examine our measurement setup rather than begin by assigning fault to the supplier.

16 **Tomas:** Include our instrument's [8] _____ and the relevant results. A certificate or a suspicion is not a substitute for the evidence from either side.

17 **Grace:** Once a cause is supported, I will ask what action addresses it and how effectiveness will be checked. A promise about future lots is not enough.

18 **Tomas:** That is the purpose of [9] _____. Distinguish the action against recurrence from replacing a piece or maintaining the current hold.

19 **Grace:** For now: three hundred held, two sampled failures, cause open, no release or rework authorized. I will send that exact status to production.

20 **Tomas:** Use [10] _____ to check whether other material may be involved. Keep any wider scope qualified until the records support it.

CHECK THE GAPS

1. Incoming inspection / **2.** nonconformance / **3.** certificate of conformity / **4.** containment / **5.** disposition / **6.** release status / **7.** correction / **8.** calibration status / **9.** corrective action / **10.** lot genealogy

Why these answers fit, page 147

F Check accuracy and scope

Three hundred pieces on hold

Return to the case

Quality engineer Tomas and supply planner Grace review two incoming component lots containing 120 and 180 pieces. One sampled piece from each lot failed a specified dimensional check. Both lots are on hold under the fictional site procedure. The cause is not established, and no release or rework disposition is authorized. Production needs parts tomorrow. Grace proposes describing all 300 pieces as defective, then releasing the second lot because its supplier certificate says conforming.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which statement distinguishes correction from corrective action?

A. Correction always proves the cause is known.

B. Corrective action is complete when material is placed on hold.

C. Correction addresses the detected problem; corrective action addresses its cause to prevent recurrence.

D. Both terms mean signing a supplier certificate.

2. Which supplier update is best supported?

A. The cause remains open; provide relevant evidence while both lots retain their current hold status.

B. Your process definitely caused every piece to fail.

C. The second lot is released because its certificate is favorable.

D. Rework is authorized despite the absence of a disposition.

Answers and explanations, page 149

Listen for the distinction

Swap roles. Keep both lots held, cause open, and release or rework unapproved despite the certificate and deadline.

Be firm about status and exact about scope

Quality communication should not soften a hold into an informal preference, but it should not overstate the evidence either. Name the held population, the inspected failures, and the authorized next step separately. Production urgency can be acknowledged without becoming an unsupported release rationale.

F Say it / transfer to a new case

Held does not mean every piece failed

A lot contains 90 pieces. Two sampled pieces failed a check. The whole lot is held, and no rework or release is authorized.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

status / disposition / ninety / two

01 **Planner:** The held quantity is [1] _____.

02 **Engineer:** The stated number of failed sampled pieces is [2] _____.

03 **Planner:** The handling decision still needs authorized [3] _____.

04 **Engineer:** A production deadline does not change the release [4] _____.

CHECK THE GAPS

1. ninety / **2.** two / **3.** disposition / **4.** status

Why these answers fit, page 149

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Three hundred pieces on hold". Grace: "The supplier offers to replace the two failed samples. That might address those particular pieces, but does not explain the dimensional problem." Tomas: [missing response] Grace: "I will request their manufacturing and inspection records. We should also examine our measurement setup rather than begin by assigning fault to the supplier." Which response belongs in the gap?

A. The certificate of conformity does not override contrary inspection evidence. That lot remains held unless the authorized process establishes a different disposition.

B. Request a disposition review. Do not tell anyone to rework or use material before the required decision, instructions, and evaluation are in place.

C. Replacement could be a correction through the approved process. It would not by itself establish the cause or prevent the same issue recurring.

D. Exactly. containment can be in place while cause remains open. Controlling the material does not require us to pretend the investigation is complete.

Answers and explanations, page 150

A Read the situation

LESSON 08 / CLINICAL TRAINING AND LABELING BOUNDARIES

Acknowledge an out-of-scope product-use request, keep the demonstration within authorized materials, and arrange a precise follow-up.

> **A useful question outside this session**
>
> Clinical trainer Sora is delivering a fictional device session using version 4 of internally approved materials for adult outpatient use. Clinician Alex asks for a pediatric home-use demonstration that the session materials do not cover. A version 3 slide remains in Sora's folder, but its current status has not been checked. No expanded-use authorization or reviewed response is supplied. Sora must acknowledge the question, avoid improvising the requested demonstration, and route it without implying that the clinician's question is improper.

People: Sora, Clinical trainer; Alex, Clinician attending the session.

Read the record

Choose one answer per question using the facts above.

1. What do the current session materials cover?

A. Every possible setting and population
B. A newly authorized expanded indication
C. Adult outpatient use, version 4
D. Pediatric home use under every condition

2. What is missing for the requested demonstration?

A. A question from a clinician
B. A previous version in the folder
C. Reviewed materials covering the requested pediatric home-use context
D. A willing audience

3. What should Sora avoid implying?

A. That the requested use is not covered by these materials
B. That the question is improper merely because it is outside the session scope
C. That version 4 is the session's current material
D. That a follow-up route is needed

Answers and explanations, page 151

B Vocabulary in use

CORE TERMS

labeling
Device-related written, printed, or graphic information accompanying or relating to the product.

review current labeling

instructions for use
The device's documented directions and relevant use information.

consult the instructions for use

indication
A specified condition or purpose for which use is described.

confirm the stated indication

contraindication
A circumstance in which the device should not be used.

identify a contraindication

warning
Information highlighting a significant hazard or safety concern.

communicate the relevant warning

precaution
Information about care needed to avoid problems during appropriate use.

explain a precaution

training scope
The users, tasks, settings, and content covered by a session.

define the training scope

approved material
Content authorized through the relevant internal review process.

use current approved materials

version control
Management of revisions so the appropriate version is used.

maintain version control

obsolete copy
A superseded document that is no longer the current controlled version.

remove an obsolete copy

demonstration unit
A device or model designated for showing functions under specified conditions.

identify the demonstration unit

simulation
A representation of a task or setting used for training or evaluation.

conduct a training simulation

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

labeling LABELING

review current labeling

Keep the meaning precise: Device-related written, printed, or graphic information accompanying or relating to the product.

precaution PRECAUTION

explain a precaution

Keep the meaning precise: Information about care needed to avoid problems during appropriate use.

B Precision and professional shorthand

EXTEND YOUR RANGE

in-service training

Instruction delivered to professionals in their work context.

schedule in-service training

competency assessment

Evaluation against defined knowledge or performance criteria.

distinguish attendance from competency assessment

attendance record

A record that a person participated in a session.

maintain an attendance record

scope boundary

The limit of what a session or role is authorized to cover.

state the scope boundary

off-label use

Use outside the product's authorized labeling, where that concept applies.

route an off-label-use question

unsolicited question

A question initiated by the recipient rather than prompted by the presenter.

document an unsolicited question

medical information request

A request routed for an appropriate reviewed scientific or product response.

route a medical information request

promotional claim

A statement used to encourage product selection or use.

review a promotional claim

disclaimer

A statement explaining a limitation or qualification.

avoid relying on a disclaimer alone

escalation route

The designated path for a question beyond the current role or scope.

follow the escalation route

B Precision and professional shorthand / continued

follow-up commitment

A specified promise about the next communication or action.

make a realistic follow-up commitment

teach-back

Asking a learner to restate or demonstrate key information to check understanding.

use teach-back appropriately

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

simulation SIMULATION

conduct a training simulation

Keep the meaning precise: A representation of a task or setting used for training or evaluation.

off-label use OFF-LABEL USE

route an off-label-use question

Keep the meaning precise: Use outside the product's authorized labeling, where that concept applies.

C Phrases that move the work forward

Acknowledge the question

That is a specific and useful question about a different population and setting.

Name the boundary

These materials do not cover pediatric home use.

Avoid judging the question

The limitation is the scope of my materials, not your ability to ask.

Keep the request precise

I will preserve the population and setting you asked about.

Separate approval types

Internal material approval does not create an expanded product authorization.

Respect clinical roles

I am not making a patient-specific treatment recommendation.

Bound the promise

I can confirm that the request was routed; I cannot promise a particular answer.

Check understanding

Please restate which use context today's session covers.

State the session scope

Today we are covering adult outpatient use with version 4 materials.

Decline the demonstration

I cannot improvise that demonstration within this session.

Offer the next route

I can document the question for the appropriate reviewed response.

Check the source

I need to confirm the current controlled version before using that slide.

Avoid a disclaimer shortcut

Adding a disclaimer would not resolve the missing reviewed content.

Clarify records

Session attendance is not the same as a completed competency assessment.

Keep the demonstration aligned

We will continue with the functions covered by the current session materials.

Close respectfully

I will follow up through the designated route while we keep this session within scope.

Precision in real exchanges

Purpose: Decline the demonstration
Useful wording: I cannot improvise that demonstration within this session.

Purpose: Avoid judging the question
Useful wording: The limitation is the scope of my materials, not your ability to ask.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which reply is best supported?

A. The question is improper and must not be recorded.

B. Pediatric home use is authorized because a clinician asked about it.

C. I can demonstrate anything after adding a disclaimer.

D. These materials cover adult outpatient use; I can route your pediatric home-use question for a reviewed response.

2. How should Sora handle the version 3 slide?

A. Assume it remains current because it is in the folder.

B. Treat its existence as expanded-use authorization.

C. Present it and correct any discrepancy after the session.

D. Check its status against the controlled source before using it.

3. A colleague needs the term for this meaning: A circumstance in which the device should not be used. Which term fits?

A. attendance record

B. version control

C. disclaimer

D. contraindication

D Choose the response that fits / continued

4. Rebuild the exchange from "A useful question outside this session". Alex: "Would an older slide help? A previous presentation mentioned another setting, though I am unsure whether it covered this device version." Sora: [missing response] Alex: "That makes sense. Please do not use an uncertain slide simply to answer immediately. A reviewed response would be more useful than a demonstration built from partial information." Which response belongs in the gap?

A. I have an older file, but its version control status has not been checked. A slide in my folder is not enough to establish that it is current or relevant.

B. I will retain that distinction in the medical information request and follow the appropriate process. We do not need to add patient-specific details that are unnecessary for this product-level question.

C. I will make that follow-up commitment through the designated process. I can confirm the action and next contact, but I cannot promise that the reviewed response will endorse the requested use.

D. I will use only the current approved material for this session. Internal approval of those materials also should not be confused with an expanded regulatory authorization for the product.

Answers and explanations, page 151

> **One useful distinction**
>
> Internally approved training material is not the same thing as a new regulatory authorization. A version number in a folder does not establish that the document is current. Confirm status through the controlled source rather than using an unchecked older slide.

E Conversation

A useful question outside this session

A useful question outside this session

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

follow-up commitment / medical information request / disclaimer / competency assessment / training scope / version control / escalation route / attendance record / unsolicited question / approved material

01 **Alex:** Could you demonstrate pediatric home use? Our team has a question about that setting, although today's examples have been outpatient adults.

02 **Sora:** I can document the question, but today's [1] _____ is adult outpatient use. The version four materials do not cover the pediatric home-use demonstration you are requesting.

03 **Alex:** I understand the session limit. I am not asking you to make a decision for an individual patient; I want to know where the product question can be addressed.

04 **Sora:** Thank you. I can use the designated [2] _____ for an appropriate reviewed response. I should preserve your specific population and setting rather than turn the request into a generic product question.

05 **Alex:** Would an older slide help? A previous presentation mentioned another setting, though I am unsure whether it covered this device version.

06 **Sora:** I have an older file, but its [3] _____ status has not been checked. A slide in my folder is not enough to establish that it is current or relevant.

E Conversation / continued

07 **Alex:** That makes sense. Please do not use an uncertain slide simply to answer immediately. A reviewed response would be more useful than a demonstration built from partial information.

08 **Sora:** I will use only the current [4] _____ for this session. Internal approval of those materials also should not be confused with an expanded regulatory authorization for the product.

09 **Alex:** Could you still show the requested setup with a disclaimer that it is outside today's scope? I want to understand whether that would resolve your concern.

10 **Sora:** A [5] _____ would not supply the missing reviewed content. I cannot improvise the demonstration and then rely on a sentence saying it is outside the materials.

Conversation continues on the next page.

A useful question outside this session

CONVERSATION / CONTINUED

11 **Alex:** Understood. Please do not imply that my question was inappropriate. We need to be able to ask about uses outside a session's scope.

12 **Sora:** Absolutely. An [6] _____ can be documented respectfully. The boundary concerns what I can demonstrate here, not your ability to raise a specific scientific or product question.

13 **Alex:** Please keep pediatric home use in the request. A general adult outpatient answer would not address the population and setting our team is asking about.

14 **Sora:** I will retain that distinction in the [7] _____ and follow the appropriate process. We do not need to add patient-specific details that are unnecessary for this product-level question.

15 **Alex:** Thank you. While we stay with today's materials, could you clarify what the attendance certificate represents? Our team should not confuse participation with a broader qualification.

16 **Sora:** The [8] _____ confirms participation. It does not by itself establish that someone has met every competency criterion or is authorized for every possible clinical use.

17 **Alex:** Then competence requires the relevant assessment; a name on the session list does not demonstrate it.

18 **Sora:** Correct. A [9] _____ has defined criteria. We should describe exactly what was assessed and avoid extending that result to populations or tasks outside the assessment.

A useful question outside this session / continued

19 **Alex:** Please route that question and tell me how we will hear back. We can continue with today's adult outpatient material while the separate response is pending.

20 **Sora:** I will make that [10] _____ through the designated process. I can confirm the action and next contact, but I cannot promise that the reviewed response will endorse the requested use.

CHECK THE GAPS

1. training scope / **2.** escalation route / **3.** version control / **4.** approved material / **5.** disclaimer / **6.** unsolicited question / **7.** medical information request / **8.** attendance record / **9.** competency assessment / **10.** follow-up commitment

Why these answers fit, page 153

F Check accuracy and scope

A useful question outside this session

Return to the case

Clinical trainer Sora is delivering a fictional device session using version 4 of internally approved materials for adult outpatient use. Clinician Alex asks for a pediatric home-use demonstration that the session materials do not cover. A version 3 slide remains in Sora's folder, but its current status has not been checked. No expanded-use authorization or reviewed response is supplied. Sora must acknowledge the question, avoid improvising the requested demonstration, and route it without implying that the clinician's question is improper.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. The attendance sheet is complete after the session. What does that establish?

A. Recorded participation in this session, not performance against competency criteria

B. Competence for every task mentioned by a participant during questions

C. Authorization for pediatric home use because the topic was discussed

D. Completion of a skills assessment for all functions on the demonstration unit

2. Which follow-up commitment is appropriate?

A. I will supply an improvised protocol tonight.

B. The reviewed response will certainly endorse the request.

C. I will route the exact question and confirm the next communication through the designated process.

D. I guarantee approval of the requested use.

Answers and explanations, page 154

Listen for the distinction

Swap roles. Route the request without improvising a demonstration or treating the question as improper.

A boundary can include a next step

A useful response names what you can cover now and what needs another route. Do not shame the questioner or pretend the question was answered by a disclaimer. Keep product training, a clinician's independent judgment, and the organization's approved response process distinct.

F Say it / transfer to a new case

Stay specific when routing a question

A session covers adult clinic use. A participant asks about pediatric home use. No reviewed material for that request is available, and the trainer will route it.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

response / home / clinic / authorized

01 **Trainer:** Today's approved materials cover adult [1] _____ use.

02 **Participant:** My question concerns pediatric use at [2] _____.

03 **Trainer:** I will route the exact request for a reviewed [3] _____.

04 **Participant:** That does not mean the requested use is already [4] _____.

CHECK THE GAPS

1. clinic / **2.** home / **3.** response / **4.** authorized

Why these answers fit, page 154

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A useful question outside this session". Alex: "Please keep pediatric home use in the request. A general adult outpatient answer would not address the population and setting our team is asking about." Sora: [missing response] Alex: "Thank you. While we stay with today's materials, could you clarify what the attendance certificate represents? Our team should not confuse participation with a broader qualification." Which response belongs in the gap?

A. I have an older file, but its version control status has not been checked. A slide in my folder is not enough to establish that it is current or relevant.

B. I will retain that distinction in the medical information request and follow the appropriate process. We do not need to add patient-specific details that are unnecessary for this product-level question.

C. I will use only the current approved material for this session. Internal approval of those materials also should not be confused with an expanded regulatory authorization for the product.

D. I will make that follow-up commitment through the designated process. I can confirm the action and next contact, but I cannot promise that the reviewed response will endorse the requested use.

Answers and explanations, page 155

Review 4 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Manufacturing, Suppliers, and Nonconformance" and "Clinical Training and Labeling Boundaries". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Material identified as not meeting a specified requirement. Which term fits?

A. nonconforming material

B. CAPA

C. supplier qualification

D. scrap

2. A colleague needs the term for this meaning: Content authorized through the relevant internal review process. Which term fits?

A. labeling

B. medical information request

C. approved material

D. in-service training

3. Rebuild the exchange from "Incomplete information still needs a handoff". Dev: "Is this automatically reportable because the display stopped? Service needs to avoid delay without deciding beyond its role." Nora: [missing response] Dev: "I will document the missing questions and the calls we make to obtain answers. The case should show what we tried, not simply remain blank." Which response belongs in the gap?

A. We should route the available event narrative for prompt internal assessment and continue follow-up. An incomplete account is not a reason to withhold it from the responsible team.

B. Seek the Unique Device Identifier and other relevant identification details through the approved process. Record what is available rather than inventing a serial number to complete the form.

C. Each follow-up attempt helps make the information history clear. New facts should be linked to their source, including when they became available and any earlier uncertainty they resolve.

D. The responsible team must assess reportability under the applicable requirements. Preserve the account and route it promptly; do not replace the assessment with an automatic yes or no.

Review 4 / Meaning and context / continued

4. Rebuild the exchange from "A useful question outside this session". Alex: "Then competence requires the relevant assessment; a name on the session list does not demonstrate it." Sora: [missing response] Alex: "Please route that question and tell me how we will hear back. We can continue with today's adult outpatient material while the separate response is pending." Which response belongs in the gap?

A. Correct. A competency assessment has defined criteria. We should describe exactly what was assessed and avoid extending that result to populations or tasks outside the assessment.

B. A disclaimer would not supply the missing reviewed content. I cannot improvise the demonstration and then rely on a sentence saying it is outside the materials.

C. The attendance record confirms participation. It does not by itself establish that someone has met every competency criterion or is authorized for every possible clinical use.

D. I can document the question, but today's training scope is adult outpatient use. The version four materials do not cover the pediatric home-use demonstration you are requesting.

Answers and explanations, page 165

Language for the next case

A small software patch still needs an impact assessment

change-impact assessment

The change-impact assessment examines the consequences of the modification, including affected functions, risks, evidence, and regulatory questions.

Ari: Not yet. The change-impact assessment is incomplete. A passing defect test covers one question, not the whole release.

regression testing

Regression testing examines whether the change adversely affects behavior that should still work, beyond the targeted defect correction.

Ari: The justified regression testing and other required checks, tied to the affected functions and risk review.

configuration baseline

The configuration baseline identifies the controlled hardware and software versions to which the evidence applies.

Ari: Correct. Confirm the configuration baseline for every relevant report so the release package refers to what was actually tested.

submission assessment

The submission assessment determines the regulatory implications of the actual change; its outcome is not established by calling the change a bug fix.

Ari: Neither blanket statement is appropriate. Regulatory affairs must document the submission assessment for this device and change.

deployment plan

The deployment plan defines the authorized update process, affected configurations, communication, and confirmation of implementation.

Ari: Yes. The deployment plan needs the affected configurations, customer instructions, and confirmation steps reviewed.

recovery procedure

The recovery procedure defines the assessed response to an unsuccessful update; it must not be improvised by customers or support staff.

Ari: The relevant recovery procedure and support arrangements must be established through the controlled process.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

A small software patch still needs an impact assessment

A device team has corrected a software defect in a test build. The fix passes its targeted test, but wider regression work and regulatory assessment are incomplete. The product manager wants to send the update to customers. No release is authorized.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

change-impact assessment / submission assessment / configuration baseline / deployment plan / recovery procedure / regression testing

01 **Sofia:** The targeted test passes. Can we send the patch to customers this afternoon and finish the remaining paperwork afterward?

02 **Ari:** Not yet. The [1] _____ is incomplete. A passing defect test covers one question, not the whole release.

03 **Sofia:** It is only a few lines of code. I thought that made it a minor maintenance change.

04 **Ari:** Code size does not establish clinical or functional impact. We need to trace the changed behavior and anything that depends on it.

05 **Sofia:** Which evidence should I expect besides repeating the test that originally exposed the defect?

06 **Ari:** The justified [2] _____ and other required checks, tied to the affected functions and risk review.

07 **Sofia:** The dashboard still shows green results from the previous version. Those should not appear to cover the new build automatically.

08 **Ari:** Correct. Confirm the [3] _____ for every relevant report so the release package refers to what was actually tested.

09 **Sofia:** Does any software change require a new submission, or can we assume a bug fix never does?

10 **Ari:** Neither blanket statement is appropriate. Regulatory affairs must document the [4] _____ for this device and change.

Conversation continues on the next page.

A small software patch still needs an impact assessment

CONVERSATION / CONTINUED

11 **Sofia:** We should also consider what customers will need to do and how the update reaches the installed devices.

12 **Ari:** Yes. The [5] _____ needs the affected configurations, customer instructions, and confirmation steps reviewed.

13 **Sofia:** Support proposed a simple message saying the update has no effect on existing workflows. We have not evaluated that claim yet.

14 **Ari:** Then do not include it as fact. Explain the verified change and any reviewed user implications without adding reassurance the evidence does not support.

15 **Sofia:** What happens if installation does not complete? Customers will need an approved route for that situation.

16 **Ari:** The relevant [6] _____ and support arrangements must be established through the controlled process.

17 **Sofia:** I will change today's milestone from customer release to completion of the review package. The targeted test result can stay visible.

18 **Ari:** That is accurate. Keep the open regression and regulatory items beside it, with owners and the evidence needed to close them.

19 **Sofia:** Once those are complete, I will ask for the authorized release decision rather than infer approval from a green dashboard.

20 **Ari:** And the customer communication must identify the approved version. A corrected defect, a completed review, and an installed update are different milestones.

CHECK THE GAPS

1. change-impact assessment / **2.** regression testing / **3.** configuration baseline / **4.** submission assessment / **5.** deployment plan / **6.** recovery procedure

Why these answers fit, page 168

Old test results are attached to a new build

A release package links test results from build 4.1 to a modified build 4.2 without an impact assessment.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

version / change / package / recorded

01 **Engineer:** These results identify build 4.1, not the modified [1] _____.

02 **Manager:** Assess which conclusions remain applicable after the [2] _____.

03 **Engineer:** Keep the open work visible in the review [3] _____.

04 **Manager:** Do not communicate release until the authorized decision is [4] _____.

CHECK THE GAPS

1. version / **2.** change / **3.** package / **4.** recorded

Why these answers fit, page 168

Keep the distinction audible

Explain the evidence and decisions required before a device-software update.

Language for the next case

A field notice reaches the hospital, but not every affected unit

affected population

The affected population is the set of devices within the notice's specified scope, not simply the number of customer emails sent.

Lena: It can show notifications sent. We still need the affected population reconciled to the model and serial-number range in the approved notice.

distribution records

Distribution records identify where devices were supplied and support follow-up when their current locations need confirmation.

Lena: Then use the distribution records to begin the trace, followed by the required customer confirmation.

Acknowledgment

Acknowledgment confirms receipt or recognition of the communication; it does not establish that every required device action has been completed.

Lena: Exactly. Acknowledgment is a communication milestone, not proof that the instructions were carried out.

device-level reconciliation

Device-level reconciliation accounts for each relevant device and its action status rather than relying on a customer-level email total.

Lena: Yes. That is device-level reconciliation. Keep unknown locations and unanswered requests visible as open items.

effectiveness checks

Effectiveness checks assess whether the field communication and required response achieved their intended purpose, rather than merely counting messages.

Lena: The plan includes effectiveness checks. Their results must be evaluated, not replaced by the number of emails in the sent folder.

closure

Closure is the authorized conclusion of the field action after its required evidence and conditions are satisfied.

Lena: Good. Any formal closure must follow the responsible team's review and applicable requirements.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

A field notice reaches the hospital, but not every affected unit

An approved fictional field notice identifies a particular model and serial-number range. A distributor has emailed ten hospitals, but no device-level responses have been received. The field-action coordinator and service lead must establish implementation status under the authorized plan. They are not determining the notice's clinical instructions or regulatory classification.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

distribution records / effectiveness checks / device-level reconciliation / Acknowledgment / closure / affected population

01 **Omar:** The distributor emailed all ten hospitals. Can the dashboard show the field action as complete?

02 **Lena:** It can show notifications sent. We still need the [1] _____ reconciled to the model and serial-number range in the approved notice.

03 **Omar:** Some hospitals have several units, and devices may have moved between departments since purchase.

04 **Lena:** Then use the [2] _____ to begin the trace, followed by the required customer confirmation.

05 **Omar:** The hospital contact may acknowledge the email without checking every unit. We need that distinction in the response form.

06 **Lena:** Exactly. [3] _____ is a communication milestone, not proof that the instructions were carried out.

Conversation / continued

07 **Omar:** One customer asks whether an older unit of the same model is included. Its serial number is outside the notice's range.

08 **Lena:** Check the exact identifier against the approved scope. Do not expand or narrow the notice casually; route any ambiguity to the responsible team.

09 **Omar:** Another contact asks what to do with a device currently in clinical use. I should not improvise a recommendation during the service call.

10 **Lena:** Use the approved instructions and the designated clinical or technical escalation route. Our coordination must not alter the authorized message.

Conversation continues on the next page.

A field notice reaches the hospital, but not every affected unit

CONVERSATION / CONTINUED

11 **Omar:** For each affected unit, we should capture the action status and the supporting confirmation required by the plan.

12 **Lena:** Yes. That is [4] _____. Keep unknown locations and unanswered requests visible as open items.

13 **Omar:** Can a nonresponse be treated as no affected stock? The distributor says silence usually means there is nothing to report.

14 **Lena:** No. Silence is not a verified inventory result. Follow the plan's contact and escalation steps and record the attempts accurately.

15 **Omar:** How do we judge whether the notification process itself has worked?

16 **Lena:** The plan includes [5] _____. Their results must be evaluated, not replaced by the number of emails in the sent folder.

17 **Omar:** I will keep receipt, device identification, and action completion as separate fields in the worklist.

18 **Lena:** Good. Any formal [6] _____ must follow the responsible team's review and applicable requirements.

19 **Omar:** Today's update will say ten hospitals contacted, responses pending, and no device-level completion yet confirmed.

20 **Lena:** That gives us a reliable starting point for follow-up without suggesting the safety communication has reached every device or been fully implemented.

CHECK THE GAPS

1. affected population / **2.** distribution records / **3.** Acknowledgment / **4.** device-level reconciliation / **5.** effectiveness checks / **6.** closure

Why these answers fit, page 170

One hospital reply is counted as six completed devices

A hospital with six potentially affected devices acknowledges a notice. Its reply does not identify the devices or confirm the required action.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

status / verified / completion / plan

01 **Service lead:** The reply confirms receipt, not device-level [1] _____.

02 **Coordinator:** Obtain the required device identifiers and action [2] _____.

03 **Service lead:** Keep the six units pending until the evidence is [3] _____.

04 **Coordinator:** Route unresolved cases through the approved follow-up [4] _____.

CHECK THE GAPS

1. completion / **2.** status / **3.** verified / **4.** plan

Why these answers fit, page 170

Keep the distinction audible

Track a device field action using identifiers and confirmed completion rather than email counts.

Language for the next case

A new carton does not automatically preserve a sterile package

packaging system

The packaging system includes the sterile barrier and relevant protective packaging; changing the outer configuration can affect protection.

Keira: Not from that fact alone. The packaging system includes how the barrier is protected during handling, transport, and storage.

sterile barrier

The sterile barrier is the packaging that maintains the sterile condition until the intended point of use, within its established conditions.

Keira: Exactly. We need an impact assessment for the sterile barrier and the product, not just a comparison of carton dimensions.

configuration

Configuration identifies the actual combination of packaging components and arrangement covered by the evidence.

Keira: The tested configuration was different. The carrier's identity does not show that the new arrangement behaves the same way.

package-integrity

Package-integrity evidence addresses whether the barrier remains intact; a photograph alone does not establish every relevant performance requirement.

Keira: Visual observations may contribute, but the plan needs justified package-integrity evidence and the other applicable performance checks.

shelf-life

Shelf-life evidence supports the claimed storage period under defined conditions; a packaging change does not automatically preserve or extend that support.

Keira: Correct. Review the shelf-life support and determine what remains applicable. Keep the claim tied to the assessed configuration and conditions.

aseptic presentation

Aseptic presentation concerns making the sterile product available at use without compromising its sterile condition, not just surviving transport.

Keira: Yes, consider aseptic presentation and the intended handling sequence through the appropriate evaluation.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

A new carton does not automatically preserve a sterile package

A device manufacturer proposes a smaller shipping carton for a terminally sterilized product. The inner sterile package is unchanged, but the fit and protective inserts will change. Existing distribution and shelf-life evidence relates to the old packaging configuration. No change is approved.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

shelf-life / sterile barrier / package-integrity / packaging system / configuration / aseptic presentation

01 **Noah:** The smaller carton reduces shipping volume, and the inner pouch is unchanged. Can we classify this as a purely cosmetic change?

02 **Keira:** Not from that fact alone. The [1] _____ includes how the barrier is protected during handling, transport, and storage.

03 **Noah:** The new insert holds the device more tightly. That could change where loads are applied to the pouch.

04 **Keira:** Exactly. We need an impact assessment for the [2] _____ and the product, not just a comparison of carton dimensions.

05 **Noah:** We already have a transport report. Why would its conclusions need review if the same carrier handles the shipment?

06 **Keira:** The tested [3] _____ was different. The carrier's identity does not show that the new arrangement behaves the same way.

Conversation / continued

07 **Noah:** Would a photograph after a trial shipment be enough to show that the pouch still looks intact?

08 **Keira:** Visual observations may contribute, but the plan needs justified [4] _____ evidence and the other applicable performance checks.

09 **Noah:** The commercial team also wants to keep the same shelf-life claim. The carton change should not become an automatic extension.

10 **Keira:** Correct. Review the [5] _____ support and determine what remains applicable. Keep the claim tied to the assessed configuration and conditions.

Conversation continues on the next page.

A new carton does not automatically preserve a sterile package

CONVERSATION / CONTINUED

11 **Noah:** Could a passing seal-strength result cover everything? It would be easier to explain one number than several reports.

12 **Keira:** No single number automatically covers the whole system. Seal strength, integrity, product protection, and opening performance answer related but different questions.

13 **Noah:** Opening matters too. A tighter insert could make removal harder even if transport protection improves.

14 **Keira:** Yes, consider [6] _____ and the intended handling sequence through the appropriate evaluation.

15 **Noah:** I will tell procurement this is a proposal under review. They should not replace the approved carton merely because the new one is available.

16 **Keira:** Good. We also need the exact component specifications and suppliers tied to the change record so the evaluation matches what would be purchased.

17 **Noah:** Can you give the review team a list of existing evidence, the gaps, and the proposed basis for any additional work?

18 **Keira:** Yes. We will justify the plan rather than automatically discard every earlier result or assume it all transfers unchanged.

19 **Noah:** Then the savings case will remain conditional on the quality, performance, and regulatory review required for the change.

20 **Keira:** That is the right status. Smaller shipping volume is a business benefit, not evidence that the revised packaging protects sterility and function.

CHECK THE GAPS

1. packaging system / **2.** sterile barrier / **3.** configuration / **4.** package-integrity / **5.** shelf-life / **6.** aseptic presentation

Why these answers fit, page 172

A packaging test covers an earlier insert

A distribution-test report identifies insert A. The proposed package uses insert B, and no impact assessment has been completed.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

conclusion / decision / configuration / reviewed

01 **Engineer:** The report covers the earlier packaging [1] _____.

02 **Manager:** Assess the effect of the new insert before reusing the [2] _____.

03 **Engineer:** Keep the change pending while the required evidence is [3] _____.

04 **Manager:** Do not substitute components before the authorized change [4] _____.

CHECK THE GAPS

1. configuration / **2.** conclusion / **3.** reviewed / **4.** decision

Why these answers fit, page 172

Keep the distinction audible

Explain packaging-system evidence and distinguish barrier integrity from a cosmetic change.

01 / Answers and explanations

A / Read the situation

A1. Who is the agreed initial audience?

B. Trained clinic staff

The brief names trained clinic staff and treats home caregivers as an unassessed expansion.

Return to the exercise, page 4

A2. What is the status of the two-minute target?

A. Proposed, not reviewed or approved

The scenario explicitly describes an unreviewed proposal rather than an established criterion.

Return to the exercise, page 4

A3. What additional evidence question follows from adding home caregivers to the brief?

D. Whether the new users, training, tasks, and home conditions support the proposed use

A new user group and setting can change how the device is used. Existing clinic scope and a proposed time target do not establish performance for home caregivers.

Return to the exercise, page 4

D / Response choices

D1. Which revision preserves the current scope?

B. Design for trained clinic staff; assess home-caregiver use as a proposed expansion.

The revision names the agreed audience and does not promote an unassessed expansion into fact.

Return to the exercise, page 8

D2. Which statement accurately describes the time target?

A. Two minutes is proposed; task boundaries and the criterion basis need review.

A number can be measurable while still lacking scope, justification, and approval.

Return to the exercise, page 8

01 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The person or user group the device is designed for. Which term fits?

B. intended user

A Not here. design assumption: An unconfirmed premise influencing a design decision. The usual combination is "challenge a design assumption".

B Correct. intended user: The person or user group the device is designed for. The usual combination is "identify intended users".

C Not here. use scenario: A description of users performing tasks under specified conditions. The usual combination is "map a use scenario".

D Not here. requirements baseline: The agreed requirement set used as a reference for controlled work. The usual combination is "approve a requirements baseline".

Return to the exercise, page 8

D4. Rebuild the exchange from "What does easy to use mean?". Owen: "Could we write setup within two minutes? It sounds clearer than easy to use, and the number would give engineering something definite to work toward." Mira: [missing response] Owen: "That distinction matters. I do not want the meeting notes to turn a suggested number into a promise that the product has already been shown to meet." Which response belongs in the gap?

D. It is a candidate acceptance criterion, not an approved one. We must define what setup includes and why two minutes is appropriate for those users.

A Not here. This line belongs to turn 8, after Owen says: "That distinction matters. I do not want the meeting notes to turn a suggested number into a promise that the product has already been shown to meet." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 12, after Owen says: "Engineering wants a simpler screen. Is that the requirement, or their proposed way to meet it? I want the brief to distinguish the two." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Owen says: "The customer summary says the device should be easy to use. I agree with that direction, but the design team says it cannot work from that sentence alone." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Mira's response in turn 6. It connects Owen's preceding point with Owen's follow-up. The time target could become a decision criterion, but its scope and basis require review first.

Return to the exercise, page 8

E / Conversation

1. user need

Turn 2. The user need preserves the concern before the team develops a more specific requirement.

01 / Answers and explanations / continued

2. scope expansion

Turn 4. Adding an unassessed user group expands the project scope rather than confirming the current design's suitability.

3. acceptance criterion

Turn 6. The time target could become a decision criterion, but its scope and basis require review first.

4. use environment

Turn 8. The use environment covers conditions such as lighting and noise that may affect the task.

5. user profile

Turn 10. The user profile identifies relevant characteristics rather than assuming different audiences have equivalent capabilities or training.

6. design output

Turn 12. A screen layout is a result of design work, not the underlying user requirement.

7. traceability

Turn 14. Traceability links the need, requirement, design result, and supporting evidence across the development records.

8. requirement owner

Turn 16. The requirement owner coordinates clarification and maintenance of the requirement without replacing specialist review.

9. requirements baseline

Turn 18. The baseline is the agreed reference set, so unreviewed proposals must not silently become approved requirements.

10. design input

Turn 20. The design input translates a defined need into a specified requirement that can guide and be evaluated in development.

Return to the conversation, page 10

01 / Answers and explanations / continued

F / Accuracy and scope

F1. What best distinguishes a need from an output?

D. The need describes the user requirement; the screen layout is a design result.

A design result can address a need but does not replace the need or evidence of fulfillment.

Return to the exercise, page 12

F2. Which meeting record is most accurate?

C. Clinic scope agreed; home-use assessment and setup criterion remain open.

The supplied discussion leaves two items unresolved and provides neither approval nor a failure result.

Return to the exercise, page 12

F / Guided speaking

1. clinic

Lead: The agreed setting is the outpatient clinic.

Turn 1. The current brief covers outpatient staff rather than the proposed home setting.

2. expansion

Engineer: Home use is a proposed expansion.

Turn 2. The additional environment broadens the scope and has not been assessed.

3. criterion

Lead: Ninety seconds is not yet an approved criterion.

Turn 3. The proposed time has not completed the stated approval process.

4. baseline

Engineer: Keep the proposal separate from the current baseline.

Turn 4. The agreed requirements reference should not silently incorporate an unapproved proposal.

Return to the conversation, page 13

01 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "What does easy to use mean?". Owen: "Then we need a record that connects the customer concern to a reviewable input and eventually to evidence. Otherwise the same question will return at every review." Mira: [missing response] Owen: "Who should resolve the setup definition? I can bring customer context, but I should not independently decide the engineering criterion or the relevant evaluation conditions." Which response belongs in the gap?

D. That is traceability. Each link should make the reasoning visible, including which assumptions remain open and which version the team has agreed.

A Not here. This line belongs to turn 4, after Owen says: "The initial audience is trained clinic staff. The latest slide adds home caregivers because sales sees a possible opportunity, although that group has not been assessed." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Owen says: "The customer summary says the device should be easy to use. I agree with that direction, but the design team says it cannot work from that sentence alone." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 16, after Owen says: "Who should resolve the setup definition? I can bring customer context, but I should not independently decide the engineering criterion or the relevant evaluation conditions." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Mira's response in turn 14. It connects Owen's preceding point with Owen's follow-up. Traceability links the need, requirement, design result, and supporting evidence across the development records.

Return to the exercise, page 13

02 / Answers and explanations

A / Read the situation

A1. Which item is the potential source of harm?

C. The accessible hot surface

The surface is the potential source, while the skin injury is a possible resulting harm.

Return to the exercise, page 14

A2. What is the status of the insulating cover?

C. Proposed, with no verified control result supplied

The briefing gives a proposed control but no implementation or effectiveness evidence.

Return to the exercise, page 14

A3. Why is low risk not established?

B. Neither a likelihood estimate nor a verified control result is supplied.

The supplied information does not support a complete risk conclusion or an effective-control claim.

Return to the exercise, page 14

D / Response choices

D1. Which chain is correctly labeled?

D. Hot surface: hazard; prolonged contact: exposure situation; burn: harm.

The chain distinguishes the potential source, the exposure circumstances, and the possible injury.

Return to the exercise, page 18

D2. Which sentence accurately reports the complaint history?

D. No complaints have been received; this alone does not establish the likelihood of harm.

Complaint absence alone cannot supply the missing probability or control-effectiveness evidence.

Return to the exercise, page 18

02 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The linked events that can lead to exposure and harm. Which term fits?

C. sequence of events

A Not here. control effectiveness: The demonstrated ability of a measure to reduce the identified risk. The usual combination is "verify control effectiveness".

B Not here. risk evaluation: Comparing estimated risk with established acceptability criteria. The usual combination is "perform a risk evaluation".

C Correct. sequence of events: The linked events that can lead to exposure and harm. The usual combination is "map the sequence of events".

D Not here. information for safety: Warnings or instructions intended to support risk reduction. The usual combination is "review information for safety".

Return to the exercise, page 18

D4. Rebuild the exchange from "The hot surface is not the injury". Sam: "Then the burn belongs in the outcome field. We have not established that it happened, so the wording must not imply an observed patient event." Elena: [missing response] Sam: "The team chose low because the complaint log is empty. That seems reassuring, but I cannot find any estimate showing how the exposure might occur." Which response belongs in the gap?

B. Correct. It is possible harm. State the outcome conditionally and preserve the distinction between this design concern and a documented incident.

A Not here. This line belongs to turn 10, after Sam says: "Should we remove the severity entry as well? It seems difficult to judge the risk while the likelihood information remains incomplete in this version." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Elena's response in turn 6. It connects Sam's preceding point with Sam's follow-up. Harm identifies the possible injury, not evidence that an injury has actually occurred.

C Not here. This line belongs to turn 18, after Sam says: "I will change the status to evidence incomplete and retain the proposed cover as an action for engineering review. We also need the criterion behind any acceptance decision." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Sam says: "The worksheet calls the skin burn a hazard and rates the issue low because we have no complaints. Can we close it with the proposed insulating cover?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 18

E / Conversation

1. hazard

Turn 2. The surface is the potential source of harm, whereas a burn is the resulting injury.

02 / Answers and explanations / continued

2. hazardous situation

Turn 4. Prolonged contact describes circumstances of exposure to the hot surface rather than the injury itself.

3. harm

Turn 6. Harm identifies the possible injury, not evidence that an injury has actually occurred.

4. probability of harm

Turn 8. The missing likelihood assessment cannot be replaced by an empty complaint log.

5. severity

Turn 10. Severity concerns the degree of possible injury and remains distinct from the probability assessment.

6. risk control

Turn 12. The cover is a proposed risk-reduction measure, not yet a demonstrated solution.

7. newly introduced risk

Turn 14. A design modification can create or change other risks, which must be considered alongside its intended benefit.

8. Residual risk

Turn 16. Residual risk is assessed after implemented controls, not assumed from an untested proposal.

9. risk acceptability

Turn 18. Acceptability depends on the established criteria and supporting evaluation, not a reassuring adjective.

10. objective evidence

Turn 20. Objective evidence is verifiable support for the eventual decision, which the current record must identify and obtain.

Return to the conversation, page 20

02 / Answers and explanations / continued

F / Accuracy and scope

F1. Which review would support a control claim for the proposed insulating cover?

C. Its implementation, effect on contact risk, and any risk introduced by the change

A proposed safeguard needs evidence that it is implemented and effective, plus assessment of consequences elsewhere in the design. A drawing, name, or complaint count alone cannot establish that.

Return to the exercise, page 22

F2. Which conclusion fits the case?

D. Further evidence is needed before the stated risk conclusion is supported.

The case supplies neither a completed risk evaluation nor a verified control result.

Return to the exercise, page 22

F / Guided speaking

1. hazard

Engineer: The sharp edge is the potential hazard.

Turn 1. The edge is the source that could cause the described injury.

2. harm

Reviewer: The cut is the possible harm.

Turn 2. A cut describes the injury rather than the physical source.

3. control

Engineer: The rounded redesign is a proposed control.

Turn 3. The change is intended to reduce risk but has not been assessed.

4. evidence

Reviewer: Its effectiveness still requires evidence.

Turn 4. A proposed change cannot be treated as effective without supporting evaluation.

Return to the conversation, page 23

02 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "The hot surface is not the injury". Sam: "Could the cover trap heat somewhere else? I want the review to consider that possibility as well as reducing contact with the hot surface." Elena: [missing response] Sam: "The draft already says residual risk acceptable. That sounds premature when the proposed measure has not been implemented or assessed against the relevant criteria." Which response belongs in the gap?

A. Exactly. Assess any newly introduced risk. A measure can address one exposure while affecting another part of the design, so its consequences need a broader review.

A Correct. This is Elena's response in turn 14. It connects Sam's preceding point with Sam's follow-up. A design modification can create or change other risks, which must be considered alongside its intended benefit.

B Not here. This line belongs to turn 8, after Sam says: "The team chose low because the complaint log is empty. That seems reassuring, but I cannot find any estimate showing how the exposure might occur." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 4, after Sam says: "I see the labeling problem. The surface could be accessible during use, and the scenario includes prolonged contact. We should keep those details in separate fields." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Sam says: "The worksheet calls the skin burn a hazard and rates the issue low because we have no complaints. Can we close it with the proposed insulating cover?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 23

03 / Answers and explanations

A / Read the situation

A1. What has passed?

C. Twelve planned specification checks in the current package

The briefing limits the passed work to twelve specification checks in one package.

Return to the exercise, page 26

A2. What has not started?

B. The separate intended-user evaluation

The intended-user evaluation is explicitly unperformed in the supplied status.

Return to the exercise, page 26

A3. What does tomorrow's meeting establish now?

B. Only that a review is scheduled

Scheduling a review does not supply its conclusions or an authorization in advance.

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D / Response choices

D1. Which slide heading is supported?

A. Twelve specification checks passed; intended-user evaluation not started.

The heading retains the actual passed work and the explicitly unperformed evaluation.

Return to the exercise, page 30

D2. A software revision follows the tests. What wording is best?

C. Assess the change before assuming the earlier results cover the new version.

The impact of the change determines what previous evidence remains applicable and what further work is needed.

Return to the exercise, page 30

03 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: A documented plan describing how a test will be performed and assessed. Which term fits?

B. test protocol

A Not here. traceability matrix: A structured table linking requirements to related evidence. The usual combination is "update the traceability matrix".

B Correct. test protocol: A documented plan describing how a test will be performed and assessed. The usual combination is "approve a test protocol".

C Not here. regression testing: Rechecking affected behavior after a change. The usual combination is "plan regression testing".

D Not here. review action: A task assigned as a result of a review. The usual combination is "track review actions".

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D4. Rebuild the exchange from "Twelve checks passed, one evaluation still open". Beth: "Can we say every requirement passed? That would be shorter, although I suspect the word every would extend the claim beyond this particular set of checks." Arun: [missing response] Beth: "I also noticed that the report refers to the earlier software build. The team discussed a revision, but I do not know whether it has been implemented." Which response belongs in the gap?

A. Show the requirement coverage instead. Each result needs a clear link to its requirement, and any uncovered requirements must remain visible in the review package.

A Correct. This is Arun's response in turn 6. It connects Beth's preceding point with Beth's follow-up. Coverage identifies the requirements actually addressed by the available evidence rather than implying universal completeness.

B Not here. This line belongs to turn 2, after Beth says: "The dashboard says validation complete because all twelve checks passed. Before tomorrow's review, I want to be sure that headline means what the team thinks it means." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 4, after Beth says: "The test result is still good progress. We should not hide it, but we should identify what those twelve checks actually cover and what remains unperformed." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 18, after Beth says: "Tomorrow, each unresolved item needs an owner. But the minutes should not show those items as closed simply because someone agrees to take responsibility." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 30

E / Conversation

1. verification

Turn 2. The specification checks confirm specified requirements, which is the verification claim supported by the case.

03 / Answers and explanations / continued

2. validation

Turn 4. Validation concerns requirements for intended use, and the separate intended-user evaluation is still unperformed.

3. requirement coverage

Turn 6. Coverage identifies the requirements actually addressed by the available evidence rather than implying universal completeness.

4. configuration

Turn 8. The configuration identifies the versions and arrangement actually tested, limiting the scope of the result.

5. change impact

Turn 10. A change-impact assessment examines whether a modification affects the relevance of existing evidence or requires more work.

6. Release authorization

Turn 12. Release authorization is a distinct formal decision and is not supplied by a test result or meeting invitation.

7. evidence package

Turn 14. The evidence package organizes the actual records supporting the review rather than relying on a dashboard color.

8. test deviation

Turn 16. A deviation is a departure from the planned test, whose significance must be assessed rather than assumed.

9. Action closure

Turn 18. Closure requires the agreed completion evidence and acceptance, not merely assignment of responsibility.

10. decision record

Turn 20. The decision record captures the review's real conclusion and conditions after the review occurs.

Return to the conversation, page 32

03 / Answers and explanations / continued

F / Accuracy and scope

F1. A reviewer accepts ownership of an open test deviation. Which status can be recorded at that point?

B. Owner assigned; required assessment, completion evidence, and acceptance still outstanding

Assignment establishes who will do the work, not its technical conclusion, completion, acceptance, or release effect. Those require their own evidence and decisions.

Return to the exercise, page 35

F2. Which conclusion may Beth report before the review?

D. The evidence package is ready for the scheduled review, with the remaining work identified.

The available facts support readiness for review but do not supply its outcome or release authorization.

Return to the exercise, page 35

F / Guided speaking

1. B

Lead: Six checks passed on version B.

Turn 1. The evidence is explicitly tied to the supplied tested version.

2. pending

Engineer: The intended-use evaluation remains pending.

Turn 2. The briefing gives no completed evaluation or successful result.

3. review

Lead: Friday is the scheduled review.

Turn 3. A booked meeting identifies a review date rather than a release decision.

4. authorized

Engineer: Release has not been authorized.

Turn 4. No release decision has been made in the supplied facts.

Return to the conversation, page 36

03 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Twelve checks passed, one evaluation still open". Beth: "I will change the status to ready for review and put the unperformed intended-user evaluation beside the completed specification package. That should prevent a misleading green summary." Arun: [missing response] Beth: "There was one departure noted during execution. I do not want to describe it as a failed requirement if that is not what happened." Which response belongs in the gap?

B. Include the evidence package, not just the totals. Reviewers need the reports, relevant versions, criteria, and any departures from the planned method.

A Not here. This line belongs to turn 12, after Beth says: "The dashboard also says ready for release. I intended ready for tomorrow's meeting, but I can see that those are not interchangeable status descriptions." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Arun's response in turn 14. It connects Beth's preceding point with Beth's follow-up. The evidence package organizes the actual records supporting the review rather than relying on a dashboard color.

C Not here. This line belongs to turn 10, after Beth says: "If the revision goes ahead, which test conclusions might it affect? Please flag that question now so the release review does not inherit mismatched versions." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Beth says: "The dashboard says validation complete because all twelve checks passed. Before tomorrow's review, I want to be sure that headline means what the team thinks it means." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 36

04 / Answers and explanations

A / Read the situation

A1. What was observed?

D. Three of eight participants selected the wrong status label in the simulated task.

The observation is limited to the stated participants, label, and simulated task.

Return to the exercise, page 37

A2. Which wording adds a cause not established by the three wrong selections?

D. Participants selected incorrectly because they were careless.

The study records a behavior, not carelessness as its cause. Wrong selection does not by itself establish misreading, misunderstanding, inattention, or any single explanation.

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A3. What is the status of the larger font?

A. An untested proposed revision

The revised label has not been tested, so its effect is not established.

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D / Response choices

D1. Which summary is most accurate?

C. Three of eight participants selected the wrong label in the formative simulated task.

The accurate summary preserves the observed count and the study's scope without adding a cause or outcome.

Return to the exercise, page 41

D2. A participant can read both labels aloud but confuses their meanings. Which issue is most directly suggested?

B. Label comprehension

Reading the characters while confusing their meaning points to comprehension rather than legibility alone.

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04 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: An assessment of a sufficiently developed design against defined evaluation goals. Which term fits?

B. summative evaluation

A Not here. use difficulty: Observed trouble completing or understanding a task. The usual combination is "record a use difficulty".

B Correct. summative evaluation: An assessment of a sufficiently developed design against defined evaluation goals. The usual combination is "plan a summative evaluation".

C Not here. cognitive load: The mental effort needed to process information and perform a task. The usual combination is "reduce cognitive load".

D Not here. root-cause analysis: Investigation of underlying contributors to an observed problem. The usual combination is "conduct root-cause analysis".

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D4. Rebuild the exchange from "Three participants read the label differently". Victor: "The team wants a larger font. That is easy to implement, but I cannot yet say that small characters caused all three wrong selections." Laila: [missing response] Victor: "One person read both labels aloud accurately and still selected the wrong status. Larger letters would not necessarily address that person's interpretation." Which response belongs in the gap?

D. Legibility is one question. We also need to examine whether the words distinguish the two states clearly in the task context.

A Not here. This line belongs to turn 10, after Victor says: "Another person finished after the moderator asked a follow-up question. The results table currently says success, with no note about the intervention." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 12, after Victor says: "What about placing the status higher on the screen? That changes where attention goes, rather than only changing the text size." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Victor says: "Three of the eight participants chose the wrong status label. The draft calls them careless. That is not something I can establish from watching the selection." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Laila's response in turn 6. It connects Victor's preceding point with Victor's follow-up. Legibility concerns distinguishing text or symbols, which font size may affect without resolving meaning.

Return to the exercise, page 41

E / Conversation

1. observation note

Turn 2. An observation note records what participants did without supplying an unsupported explanation for their behavior.

04 / Answers and explanations / continued

2. formative evaluation

Turn 4. A formative evaluation informs design development and does not establish a population error rate or completed validation.

3. Legibility

Turn 6. Legibility concerns distinguishing text or symbols, which font size may affect without resolving meaning.

4. label comprehension

Turn 8. Reading text correctly while misunderstanding its meaning suggests a comprehension issue rather than only a visibility problem.

5. prompting

Turn 10. Moderator prompting can influence performance and must remain visible when interpreting task completion.

6. visual hierarchy

Turn 12. Visual hierarchy concerns the relative prominence and arrangement of information, beyond the size of individual characters.

7. design iteration

Turn 14. A design iteration is a revised version to evaluate, not evidence that the underlying problem is resolved.

8. critical task

Turn 16. Critical-task status depends on the potential harm of incorrect or omitted performance and requires the appropriate assessment.

9. human-factors validation

Turn 18. The supplied exploratory study and untested revision do not establish completed human-factors validation.

10. root-cause analysis

Turn 20. Root-cause analysis examines contributors to the observed difficulty rather than accepting an unsupported blame statement.

Return to the conversation, page 43

04 / Answers and explanations / continued

F / Accuracy and scope

F1. How should assistance during a task be reported?

B. Document the prompting rather than presenting assisted success as unassisted performance.

Prompting can affect the interpretation of performance and must remain visible in the record.

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F2. Which next-status sentence fits the facts?

D. The larger-font revision is proposed; its effect on the observed difficulty remains untested.

The supplied facts contain a proposal but no evaluation of that revised label.

Return to the exercise, page 45

F / Guided speaking

1. meaning

Researcher: Two of ten had difficulty with the label's meaning.

Turn 1. The participants could read the text but misunderstood what it conveyed.

2. session

Designer: This is an observation from a formative session.

Turn 2. The supplied result comes from a developmental evaluation, not a population study.

3. untested

Researcher: The new wording remains untested.

Turn 3. No evaluation of the proposed wording has been performed.

4. resolved

Designer: We cannot yet claim the difficulty is resolved.

Turn 4. A proposed revision does not demonstrate that the observed problem has been corrected.

Return to the conversation, page 46

04 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Three participants read the label differently". Victor: "I will stop calling the larger font a fix. The revised label has not been tested, and we may need a different approach." Laila: [missing response] Victor: "The project lead also asks whether a wrong selection makes this a critical task. Three errors alone do not answer the safety question, do they?" Which response belongs in the gap?

B. Call it a proposed design iteration. Link it to the observed difficulty and keep the follow-up result open rather than writing success in advance.

A Not here. This line belongs to turn 16, after Victor says: "The project lead also asks whether a wrong selection makes this a critical task. Three errors alone do not answer the safety question, do they?" It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Laila's response in turn 14. It connects Victor's preceding point with Victor's follow-up. A design iteration is a revised version to evaluate, not evidence that the underlying problem is resolved.

C Not here. This line belongs to turn 4, after Victor says: "Would thirty-seven point five percent sound more precise in the headline? I worry the count looks small, even though the issue may matter." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 20, after Victor says: "I will retain the three-of-eight count, the reading-versus-meaning observation, and the assisted completion. Proposed changes will sit in a separate section." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 46

05 / Answers and explanations

A / Read the situation

A1. What is the current pathway status?

B. A possible 510(k) route is being evaluated.

The briefing identifies an ongoing pathway assessment, not a completed submission or decision.

Return to the exercise, page 49

A2. What is known about the predicate?

D. A candidate has been identified, but suitability remains under review.

The candidate's identification does not settle whether it supports the proposed pathway.

Return to the exercise, page 49

A3. Why is the October launch promise unsupported?

B. No filing or regulatory decision exists, and the pathway assessment is incomplete.

The case supplies unresolved dependencies rather than a guaranteed launch authorization or date.

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D / Response choices

D1. Which sales-slide correction matches the case?

B. Possible 510(k) pathway under internal assessment; no submission or FDA decision yet.

The correction describes the actual internal stage without inventing approval, timing, or rejection.

Return to the exercise, page 53

D2. Which statement about the candidate predicate is accurate?

D. Identification begins the assessment; suitability and the comparative evidence still need review.

A named candidate is not itself an accepted comparison or a regulatory decision.

Return to the exercise, page 53

05 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: A US premarket notification seeking a substantial-equivalence determination. Which term fits?

B. 510(k)

A Not here. response package: Documents addressing specified reviewer questions or deficiencies. The usual combination is "compile a response package".

B Correct. 510(k): A US premarket notification seeking a substantial-equivalence determination. The usual combination is "prepare a 510(k)".

C Not here. clearance: A favorable 510(k) determination permitting marketing within its scope. The usual combination is "obtain 510(k) clearance".

D Not here. submission dossier: The organized documents and evidence submitted for regulatory review. The usual combination is "assemble a submission dossier".

Return to the exercise, page 53

D4. Rebuild the exchange from "A route under review is not a clearance". Caleb: "What should the commercial team understand about that comparison? I want the next discussion to focus on the evidence rather than repeat the same assumption." Inez: [missing response] Caleb: "Nothing has been filed, correct? Then under review is too vague; sales could read that as an FDA review already underway." Which response belongs in the gap?

B. The relevant determination is substantial equivalence. The proposed use and technological characteristics matter, and the evidence must support the actual comparison rather than a convenient headline.

A Not here. This line belongs to turn 12, after Caleb says: "Could a different route still be needed? I do not want the revised slide to make the 510(k) plan sound irrevocable while the assessment is incomplete." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Inez's response in turn 6. It connects Caleb's preceding point with Caleb's follow-up. Substantial equivalence is the comparative regulatory determination sought through a 510(k), not a claim established by appearance.

C Not here. This line belongs to turn 2, after Caleb says: "The launch slide says FDA approved and shows October as the launch month. I need to understand whether those statements reflect an actual decision or our working plan." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 4, after Caleb says: "The team found a marketed device that looks similar. I took that as confirmation that the route was settled, but the comparison may not be that simple." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 53

E / Conversation

1. regulatory pathway

Turn 2. The pathway is the route being evaluated, not an agency decision already obtained.

05 / Answers and explanations / continued

2. predicate device

Turn 4. A candidate predicate is a comparator under assessment, not proof that the route or outcome is settled.

3. substantial equivalence

Turn 6. Substantial equivalence is the comparative regulatory determination sought through a 510(k), not a claim established by appearance.

4. submission dossier

Turn 8. The dossier is the organized submission material, which has not yet been filed in this case.

5. clearance

Turn 10. Clearance is the appropriate favorable 510(k) decision term, distinct from PMA approval.

6. regulatory strategy

Turn 12. The strategy coordinates the applicable route and evidence; it remains subject to the unresolved assessment.

7. indications for use

Turn 14. Indications describe the specific proposed purposes and population, which must align with the regulatory assessment and evidence.

8. launch assumption

Turn 16. The October date is a conditional planning premise rather than an authorized or guaranteed launch date.

9. pre-submission interaction

Turn 18. A pre-submission interaction can provide feedback but does not itself authorize marketing.

10. marketing authorization

Turn 20. Marketing authorization is the applicable permission to market within a defined scope, which this case does not supply.

Return to the conversation, page 55

05 / Answers and explanations / continued

F / Accuracy and scope

F1. Which October-launch note preserves the current regulatory status?

C. Internal planning target; pathway and predicate still under assessment, with no filing or decision

The date is internal and conditional. Neither internal approval, a candidate comparator, nor a pathway discussion starts or settles the required agency decision.

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F2. What does a pre-submission interaction establish by itself?

C. An opportunity for feedback, not marketing authorization

A pre-submission exchange does not itself supply the later marketing decision.

Return to the exercise, page 58

F / Guided speaking

1. filed

Lead: The submission has been filed.

Turn 1. The facts establish submission but do not establish a favorable decision.

2. ongoing

Specialist: The agency's review remains ongoing.

Turn 2. The case explicitly places the submission in an unfinished review.

3. clearance

Lead: No 510(k) clearance has been issued.

Turn 3. Clearance is the relevant favorable 510(k) decision, which is not yet available.

4. assumption

Specialist: The target date is a planning assumption.

Turn 4. An internal date does not guarantee the timing or outcome of regulatory review.

Return to the conversation, page 59

05 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A route under review is not a clearance". Caleb: "The product team also wants broader wording about who can use the device. That seems connected to the pathway work, not just to marketing style." Inez: [missing response] Caleb: "Then October should be shown as an internal target with conditions. We cannot treat it as an agency commitment when neither the pathway nor the evidence package is settled." Which response belongs in the gap?

B. It is. The indications for use must be clear and consistent. A broader statement can change the claim being assessed and the evidence needed to support it.

A Not here. This line belongs to turn 12, after Caleb says: "Could a different route still be needed? I do not want the revised slide to make the 510(k) plan sound irrevocable while the assessment is incomplete." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Inez's response in turn 14. It connects Caleb's preceding point with Caleb's follow-up. Indications describe the specific proposed purposes and population, which must align with the regulatory assessment and evidence.

C Not here. This line belongs to turn 2, after Caleb says: "The launch slide says FDA approved and shows October as the launch month. I need to understand whether those statements reflect an actual decision or our working plan." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 6, after Caleb says: "What should the commercial team understand about that comparison? I want the next discussion to focus on the evidence rather than repeat the same assumption." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 59

06 / Answers and explanations

A / Read the situation

A1. Which time is known?

B. The call-receipt time, 14:10 today

The supplied time records receipt of the call, not the underlying event.

Return to the exercise, page 61

A2. What is known about patient outcome?

D. It is not yet known.

The case supplies no outcome, so neither absence nor presence of injury is established.

Return to the exercise, page 61

A3. What does eight versus four establish?

C. The complaint count doubled, but no rate can be calculated from the supplied data.

Counts can be compared directly, but a use-based rate requires a relevant denominator and context.

Return to the exercise, page 61

D / Response choices

D1. Which intake sentence preserves the facts?

B. Display interruption reported; event date, identifier, and patient outcome unknown; call received today at 14:10.

The accurate sentence distinguishes supplied facts, unknown fields, and receipt time without inventing an outcome.

Return to the exercise, page 65

D2. What is the appropriate internal handoff wording?

C. Route the available account promptly for assessment and continue documented follow-up.

Missing information should trigger follow-up rather than replace the required internal review.

Return to the exercise, page 65

06 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The US system for reporting specified device events under applicable requirements. Which term fits?

C. Medical Device Reporting

A Not here. trend analysis: Review of patterns in data over time. The usual combination is "perform trend analysis".

B Not here. receipt date: The date the organization received the communication. The usual combination is "record the receipt date".

C Correct. Medical Device Reporting: The US system for reporting specified device events under applicable requirements. The usual combination is "assess Medical Device Reporting obligations".

D Not here. event narrative: A factual account of the reported occurrence and its context. The usual combination is "preserve the event narrative".

Return to the exercise, page 65

D4. Rebuild the exchange from "Incomplete information still needs a handoff". Dev: "The caller did not tell us whether anyone was injured. The draft summary says no injury, which is more definite than the information actually supports." Nora: [missing response] Dev: "We also need to identify the unit accurately. A model name alone might not let the investigation team distinguish this device from other units at the site." Which response belongs in the gap?

D. Change patient outcome to unknown. We should ask for clarification and record the response without treating silence in the first call as a confirmed absence of harm.

A Not here. This line belongs to turn 2, after Dev says: "The caller said the monitor stopped displaying, but the device identifier and event date are missing. I thought we should wait before sending an incomplete record onward." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 18, after Dev says: "Some records might refer to the same occurrence. We should check that before treating every entry as a separate event or comparing the clinical outcomes." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 14, after Dev says: "The summary says device-caused. Where did that conclusion come from? The caller described the display stopping; we do not yet have the cause." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Nora's response in turn 6. It connects Dev's preceding point with Dev's follow-up. Patient outcome is unknown here; lack of supplied information does not establish that no injury occurred.

Return to the exercise, page 65

E / Conversation

1. event narrative

Turn 2. The event narrative preserves the reported occurrence while missing facts are pursued through follow-up.

06 / Answers and explanations / continued

2. receipt date

Turn 4. The supplied date and time concern receipt of the communication, not the unconfirmed event date.

3. patient outcome

Turn 6. Patient outcome is unknown here; lack of supplied information does not establish that no injury occurred.

4. Unique Device Identifier

Turn 8. The Unique Device Identifier helps identify the device, while unavailable identification details remain follow-up items.

5. reportability

Turn 10. Reportability depends on applicable criteria and the event assessment, not the intake team's unsupported assumption.

6. follow-up attempt

Turn 12. A follow-up attempt records efforts to obtain the information needed for the investigation and reporting assessment.

7. causal relationship

Turn 14. A causal relationship is not established merely because the caller reported an event involving the device.

8. exposure denominator

Turn 16. A rate relative to use requires the relevant exposure quantity, which the dashboard does not provide.

9. duplicate report

Turn 18. A duplicate report may describe an already recorded event, so record counts are not automatically unique-event counts.

10. supplemental report

Turn 20. A supplemental report provides required later information through the appropriate reporting process rather than silently altering history.

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06 / Answers and explanations / continued

F / Accuracy and scope

F1. What does the dashboard support?

D. The count rose from four to eight; the rate per use is not established.

The dashboard supplies counts, not exposure, unique-event confirmation, outcomes, or causation.

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F2. The caller later reports no injury. Which update preserves the information history and review responsibilities?

D. Add the attributed clarification and its receipt time; continue the applicable reporting assessment.

A later no-injury report resolves an outcome detail, not the initial knowledge state or every reporting criterion. Some malfunction circumstances can require reporting assessment even without an injury.

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F / Guided speaking

1. received

Specialist: Today is the date the report was received.

Turn 1. The case gives the communication date but not the event date.

2. unknown

Colleague: The patient outcome is still unknown.

Turn 2. No outcome was supplied, so absence of injury cannot be assumed.

3. count

Specialist: Six is the complaint count.

Turn 3. Six describes recorded complaints rather than a rate per use.

4. denominator

Colleague: A use-based rate needs the exposure denominator.

Turn 4. The relevant number of uses is missing from the supplied data.

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06 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Incomplete information still needs a handoff". Dev: "The summary says device-caused. Where did that conclusion come from? The caller described the display stopping; we do not yet have the cause." Nora: [missing response] Dev: "The quarterly dashboard adds another concern: eight complaints this quarter and four last quarter. The presentation says the rate doubled, but there is no use-volume information." Which response belongs in the gap?

A. Then the causal relationship remains unconfirmed. Report the observation accurately without either asserting device causation or dismissing the device as unrelated before the evidence supports that conclusion.

A Correct. This is Nora's response in turn 14. It connects Dev's preceding point with Dev's follow-up. A causal relationship is not established merely because the caller reported an event involving the device.

B Not here. This line belongs to turn 6, after Dev says: "The caller did not tell us whether anyone was injured. The draft summary says no injury, which is more definite than the information actually supports." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Dev says: "The caller said the monitor stopped displaying, but the device identifier and event date are missing. I thought we should wait before sending an incomplete record onward." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 18, after Dev says: "Some records might refer to the same occurrence. We should check that before treating every entry as a separate event or comparing the clinical outcomes." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 72

07 / Answers and explanations

A / Read the situation

A1. How many pieces are in the two held lots?

D. 300

The held-lot quantities are 120 and 180, which total 300 pieces.

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A2. How many sampled pieces are explicitly reported as failing?

C. Two, one from each lot

The brief reports two observed sampled failures, not inspection results for every piece.

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A3. What is the current disposition?

D. Both lots remain on hold; no release or rework is authorized.

The case supplies a hold and explicitly states that disposition has not been authorized.

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D / Response choices

D1. Which quantity statement is accurate?

C. Three hundred pieces are held; two sampled pieces are explicitly reported as failing.

The statement distinguishes total held quantity from the two observed sampled failures.

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D2. Production proposes using the 180-piece lot because its certificate is conforming. Which response preserves the current evidence and authority?

B. The failed sample and hold remain; the certificate does not replace an authorized disposition.

The hold covers the whole lot. A certificate, missing individual results, and deadline do not supply the missing handling decision or cancel contrary inspection evidence.

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07 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Immediate action to limit the scope or spread of a problem. Which term fits?

C. containment

A Not here. lot genealogy: Records linking material lots through production and distribution. The usual combination is "trace lot genealogy".

B Not here. concession: An authorized permission to use or release specified nonconforming product under defined conditions. The usual combination is "evaluate a concession request".

C Correct. containment: Immediate action to limit the scope or spread of a problem. The usual combination is "implement containment".

D Not here. change notification: Communication of a proposed or implemented change affecting supply. The usual combination is "require a change notification".

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D4. Rebuild the exchange from "Three hundred pieces on hold". Grace: "The 180-piece lot has a supplier certificate saying conforming. Can that support using it while we investigate the failed sample?" Tomas: [missing response] Grace: "I will tell production both lots remain unavailable. We have not established whether the two sampled failures have the same cause." Which response belongs in the gap?

D. The certificate of conformity does not override contrary inspection evidence. That lot remains held unless the authorized process establishes a different disposition.

A Not here. This line belongs to turn 4, after Grace says: "So the report needs three hundred held, two observed failures. We cannot assign an individual result to pieces that were not inspected." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Grace says: "Production needs the parts tomorrow. The lots contain 120 and 180 pieces. I was about to report all 300 as defective, but that overstates our inspection." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Grace says: "I will tell production both lots remain unavailable. We have not established whether the two sampled failures have the same cause." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Tomas's response in turn 6. It connects Grace's preceding point with Grace's follow-up. A certificate is a supplier statement and does not override contrary inspection evidence or the current hold.

Return to the exercise, page 81

E / Conversation

1. Incoming inspection

Turn 2. Incoming inspection produced two sampled failures; it did not establish individual results for every piece.

07 / Answers and explanations / continued

2. nonconformance

Turn 4. The nonconformance is the failure against the specified dimensional requirement, with its observed scope documented.

3. certificate of conformity

Turn 6. A certificate is a supplier statement and does not override contrary inspection evidence or the current hold.

4. containment

Turn 8. Containment limits unintended use while the scope and cause are investigated; it is not a final causal conclusion.

5. disposition

Turn 10. Disposition is the authorized handling decision, which has not yet been made for these lots.

6. release status

Turn 12. Release status records whether use is authorized, independently of schedule pressure or a hoped-for resolution.

7. correction

Turn 14. Correction addresses a detected nonconformity itself and does not necessarily resolve the underlying cause.

8. calibration status

Turn 16. Calibration status is relevant measurement information to review, not automatic proof that either party caused the problem.

9. corrective action

Turn 18. Corrective action addresses a cause to prevent recurrence, with evidence needed to support its effectiveness.

10. lot genealogy

Turn 20. Lot genealogy links material through production and distribution, supporting assessment of any wider affected scope.

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07 / Answers and explanations / continued

F / Accuracy and scope

F1. Which statement distinguishes correction from corrective action?

C. Correction addresses the detected problem; corrective action addresses its cause to prevent recurrence.

The distinction concerns fixing the detected nonconformity versus addressing the cause of recurrence.

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F2. Which supplier update is best supported?

A. The cause remains open; provide relevant evidence while both lots retain their current hold status.

The supplied evidence supports an open investigation and continued stated hold, not a cause or release decision.

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F / Guided speaking

1. ninety

Planner: The held quantity is ninety.

Turn 1. The hold covers the whole ninety-piece lot, not just the inspected failures.

2. two

Engineer: The stated number of failed sampled pieces is two.

Turn 2. Only two sampled failures are supplied in the case.

3. disposition

Planner: The handling decision still needs authorized disposition.

Turn 3. No decision permitting release or rework has been made.

4. status

Engineer: A production deadline does not change the release status.

Turn 4. Schedule urgency does not supply the missing authorization for material use.

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07 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Three hundred pieces on hold". Grace: "The supplier offers to replace the two failed samples. That might address those particular pieces, but does not explain the dimensional problem." Tomas: [missing response] Grace: "I will request their manufacturing and inspection records. We should also examine our measurement setup rather than begin by assigning fault to the supplier." Which response belongs in the gap?

C. Replacement could be a correction through the approved process. It would not by itself establish the cause or prevent the same issue recurring.

A Not here. This line belongs to turn 6, after Grace says: "The 180-piece lot has a supplier certificate saying conforming. Can that support using it while we investigate the failed sample?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 10, after Grace says: "Could I ask quality about rework tonight? I need to offer production a next step without accidentally authorizing work myself." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Tomas's response in turn 14. It connects Grace's preceding point with Grace's follow-up. Correction addresses a detected nonconformity itself and does not necessarily resolve the underlying cause.

D Not here. This line belongs to turn 8, after Grace says: "I will tell production both lots remain unavailable. We have not established whether the two sampled failures have the same cause." It addresses that point, not the missing turn in the supplied exchange.

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08 / Answers and explanations

A / Read the situation

A1. What do the current session materials cover?

C. Adult outpatient use, version 4

The brief limits the internally approved session materials to adult outpatient use in version four.

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A2. What is missing for the requested demonstration?

C. Reviewed materials covering the requested pediatric home-use context

Neither the request nor an unchecked old slide supplies the reviewed scope for the demonstration.

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A3. What should Sora avoid implying?

B. That the question is improper merely because it is outside the session scope

The scenario calls for a respectful boundary, not a judgment that asking the question is itself improper.

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D / Response choices

D1. Which reply is best supported?

D. These materials cover adult outpatient use; I can route your pediatric home-use question for a reviewed response.

The reply respects the question while accurately limiting the current materials and offering a review route.

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D2. How should Sora handle the version 3 slide?

D. Check its status against the controlled source before using it.

The folder copy has unverified status and cannot establish that the content is current.

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08 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: A circumstance in which the device should not be used. Which term fits?

D. contraindication

A Not here. attendance record: A record that a person participated in a session. The usual combination is "maintain an attendance record".

B Not here. version control: Management of revisions so the appropriate version is used. The usual combination is "maintain version control".

C Not here. disclaimer: A statement explaining a limitation or qualification. The usual combination is "avoid relying on a disclaimer alone".

D Correct. contraindication: A circumstance in which the device should not be used. The usual combination is "identify a contraindication".

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D4. Rebuild the exchange from "A useful question outside this session". Alex: "Would an older slide help? A previous presentation mentioned another setting, though I am unsure whether it covered this device version." Sora: [missing response] Alex: "That makes sense. Please do not use an uncertain slide simply to answer immediately. A reviewed response would be more useful than a demonstration built from partial information." Which response belongs in the gap?

A. I have an older file, but its version control status has not been checked. A slide in my folder is not enough to establish that it is current or relevant.

A Correct. This is Sora's response in turn 6. It connects Alex's preceding point with Alex's follow-up. Version control requires checking the valid controlled source rather than trusting an older file's presence.

B Not here. This line belongs to turn 14, after Alex says: "Please keep pediatric home use in the request. A general adult outpatient answer would not address the population and setting our team is asking about." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Alex says: "Please route that question and tell me how we will hear back. We can continue with today's adult outpatient material while the separate response is pending." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Alex says: "That makes sense. Please do not use an uncertain slide simply to answer immediately. A reviewed response would be more useful than a demonstration built from partial information." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 92

08 / Answers and explanations / continued

E / Conversation

1. training scope

Turn 2. The session scope is adult outpatient use and does not include the requested population and setting.

2. escalation route

Turn 4. The escalation route directs a question beyond the trainer's materials to the appropriate review process.

3. version control

Turn 6. Version control requires checking the valid controlled source rather than trusting an older file's presence.

4. approved material

Turn 8. Approved material means content authorized through the relevant internal process, not an expanded regulatory permission.

5. disclaimer

Turn 10. A disclaimer does not supply the missing reviewed content or make an unsupported demonstration appropriate.

6. unsolicited question

Turn 12. The clinician initiated the question; recording it does not imply that asking it was improper.

7. medical information request

Turn 14. The request preserves the exact population and setting for an appropriate reviewed response.

8. attendance record

Turn 16. An attendance record documents participation, not proof of competence across uses or clinical authority.

9. competency assessment

Turn 18. A competency assessment evaluates defined knowledge or performance criteria beyond mere attendance.

10. follow-up commitment

Turn 20. The commitment concerns routing and communication, not a guaranteed substantive answer or authorization.

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08 / Answers and explanations / continued

F / Accuracy and scope

F1. The attendance sheet is complete after the session. What does that establish?

A. Recorded participation in this session, not performance against competency criteria

Attendance records presence. Competence needs the defined assessment, while expanded-use authorization is a separate question that a signature or discussion cannot establish.

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F2. Which follow-up commitment is appropriate?

C. I will route the exact question and confirm the next communication through the designated process.

The trainer can commit to routing and communication without promising an unsupported substantive outcome.

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F / Guided speaking

1. clinic

Trainer: Today's approved materials cover adult clinic use.

Turn 1. The session scope is adult clinic use, not the requested home context.

2. home

Participant: My question concerns pediatric use at home.

Turn 2. The requested setting must remain explicit in the handoff.

3. response

Trainer: I will route the exact request for a reviewed response.

Turn 3. Routing seeks an appropriate response without inventing content in the session.

4. authorized

Participant: That does not mean the requested use is already authorized.

Turn 4. A question or referral does not itself create authorization for the requested use.

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08 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A useful question outside this session". Alex: "Please keep pediatric home use in the request. A general adult outpatient answer would not address the population and setting our team is asking about." Sora: [missing response] Alex: "Thank you. While we stay with today's materials, could you clarify what the attendance certificate represents? Our team should not confuse participation with a broader qualification." Which response belongs in the gap?

B. I will retain that distinction in the medical information request and follow the appropriate process. We do not need to add patient-specific details that are unnecessary for this product-level question.

A Not here. This line belongs to turn 6, after Alex says: "Would an older slide help? A previous presentation mentioned another setting, though I am unsure whether it covered this device version." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Sora's response in turn 14. It connects Alex's preceding point with Alex's follow-up. The request preserves the exact population and setting for an appropriate reviewed response.

C Not here. This line belongs to turn 8, after Alex says: "That makes sense. Please do not use an uncertain slide simply to answer immediately. A reviewed response would be more useful than a demonstration built from partial information." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 20, after Alex says: "Please route that question and tell me how we will hear back. We can continue with today's adult outpatient material while the separate response is pending." It addresses that point, not the missing turn in the supplied exchange.

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Review 1 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A result of design work that can be checked against inputs. Which term fits?

A. design output

A Correct. design output: A result of design work that can be checked against inputs. The usual combination is "review a design output".

B Not here. voice of the customer: Collected customer needs and expectations. The usual combination is "analyze the voice of the customer".

C Not here. task analysis: Examination of the actions and decisions needed to perform a task. The usual combination is "conduct a task analysis".

D Not here. workflow: The ordered activities through which work is completed. The usual combination is "observe the clinical workflow".

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2. A colleague needs the term for this meaning: Systematic identification of hazards and estimation of associated risks. Which term fits?

A. risk analysis

A Correct. risk analysis: Systematic identification of hazards and estimation of associated risks. The usual combination is "document the risk analysis".

B Not here. protective measure: A safeguard intended to reduce exposure or its consequences. The usual combination is "assess a protective measure".

C Not here. hazard: A potential source of injury or other harm. The usual combination is "identify a hazard".

D Not here. normal condition: Operation without a fault being present. The usual combination is "assess normal conditions".

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Review 1 / Explanations / continued

3. Rebuild the exchange from "What does easy to use mean?". Owen: "The sales slide says suitable everywhere. We have not gathered evidence from homes, so I will remove that claim from the current product description." Mira: [missing response] Owen: "Engineering wants a simpler screen. Is that the requirement, or their proposed way to meet it? I want the brief to distinguish the two." Which response belongs in the gap?

A. Good. The user profile should describe relevant training and capabilities. We cannot assume that clinic staff and home caregivers approach the same instruction in the same way.

A Correct. This is Mira's response in turn 10. It connects Owen's preceding point with Owen's follow-up. The user profile identifies relevant characteristics rather than assuming different audiences have equivalent capabilities or training.

B Not here. This line belongs to turn 2, after Owen says: "The customer summary says the device should be easy to use. I agree with that direction, but the design team says it cannot work from that sentence alone." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Owen says: "I will keep easy to use as the customer concern and assign the setup-definition work. The home-use proposal and two-minute target remain open for review." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 18, after Owen says: "I will separate the clinic requirement from the home-use proposal. The notes will say that the two-minute target remains unreviewed, with a named follow-up owner." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 24

Review 1 / Explanations / continued

4. Rebuild the exchange from "The hot surface is not the injury". Sam: "I will change the status to evidence incomplete and retain the proposed cover as an action for engineering review. We also need the criterion behind any acceptance decision." Elena: [missing response] Sam: "I will separate the hot surface, prolonged contact, and possible burn in the record. Send me the evidence needed before we can judge the proposed cover." Which response belongs in the gap?

C. Yes, document the applicable risk acceptability criteria. The reviewer needs the reasoning and evidence, not simply a low label attached to an unfinished analysis.

A Not here. This line belongs to turn 8, after Sam says: "The team chose low because the complaint log is empty. That seems reassuring, but I cannot find any estimate showing how the exposure might occur." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 10, after Sam says: "Should we remove the severity entry as well? It seems difficult to judge the risk while the likelihood information remains incomplete in this version." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Elena's response in turn 18. It connects Sam's preceding point with Sam's follow-up. Acceptability depends on the established criteria and supporting evaluation, not a reassuring adjective.

D Not here. This line belongs to turn 2, after Sam says: "The worksheet calls the skin burn a hazard and rates the issue low because we have no complaints. Can we close it with the proposed insulating cover?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 24

Review 2 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A systematic evaluation of design progress, evidence, and issues. Which term fits?

B. design review

A Not here. simulated use: Evaluation under conditions designed to represent actual use. The usual combination is "define simulated-use conditions".

B Correct. design review: A systematic evaluation of design progress, evidence, and issues. The usual combination is "hold a design review".

C Not here. requirement coverage: The extent to which requirements are addressed by specified evidence. The usual combination is "assess requirement coverage".

D Not here. change impact: The consequences of a proposed or implemented alteration. The usual combination is "assess change impact".

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2. A colleague needs the term for this meaning: A situation in which a problem nearly occurs but is avoided. Which term fits?

B. close call

A Not here. visual hierarchy: The arrangement that makes some information more prominent than other information. The usual combination is "improve the visual hierarchy".

B Correct. close call: A situation in which a problem nearly occurs but is avoided. The usual combination is "document a close call".

C Not here. task success: Completion of a defined task under stated evaluation criteria. The usual combination is "define task success".

D Not here. human factors: Study and application of how people interact with systems and products. The usual combination is "apply human-factors principles".

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Review 2 / Explanations / continued

3. Rebuild the exchange from "The hot surface is not the injury". Sam: "Should we remove the severity entry as well? It seems difficult to judge the risk while the likelihood information remains incomplete in this version." Elena: [missing response] Sam: "The cover remains attractive as an engineering proposal. I want to record it without giving the impression that it has already solved the issue." Which response belongs in the gap?

A. Keep severity separate from likelihood. We need to describe the possible injury carefully; uncertainty in one dimension does not make the other dimension disappear.

A Correct. This is Elena's response in turn 10. It connects Sam's preceding point with Sam's follow-up. Severity concerns the degree of possible injury and remains distinct from the probability assessment.

B Not here. This line belongs to turn 2, after Sam says: "The worksheet calls the skin burn a hazard and rates the issue low because we have no complaints. Can we close it with the proposed insulating cover?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Sam says: "The team chose low because the complaint log is empty. That seems reassuring, but I cannot find any estimate showing how the exposure might occur." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 18, after Sam says: "I will change the status to evidence incomplete and retain the proposed cover as an action for engineering review. We also need the criterion behind any acceptance decision." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 47

Review 2 / Explanations / continued

4. Rebuild the exchange from "Three participants read the label differently". Victor: "Then I should not describe this as our final usability validation either. We used the session to find weaknesses while the interface was still developing." Laila: [missing response] Victor: "I will retain the three-of-eight count, the reading-versus-meaning observation, and the assisted completion. Proposed changes will sit in a separate section." Which response belongs in the gap?

C. Correct. Completed human-factors validation is not established by a formative session and an untested revision. State the evaluation stage and conditions clearly.

A Not here. This line belongs to turn 6, after Victor says: "The team wants a larger font. That is easy to implement, but I cannot yet say that small characters caused all three wrong selections." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Victor says: "Would thirty-seven point five percent sound more precise in the headline? I worry the count looks small, even though the issue may matter." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Laila's response in turn 18. It connects Victor's preceding point with Victor's follow-up. The supplied exploratory study and untested revision do not establish completed human-factors validation.

D Not here. This line belongs to turn 2, after Victor says: "Three of the eight participants chose the wrong status label. The draft calls them careless. That is not something I can establish from watching the selection." It addresses that point, not the missing turn in the supplied exchange.

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Review 3 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A code used by FDA to identify a device type. Which term fits?

D. product code

A Not here. deficiency: A shortcoming identified in the submitted information. The usual combination is "respond to a deficiency".

B Not here. premarket approval: The US PMA pathway requiring an independent safety-and-effectiveness demonstration. The usual combination is "prepare a premarket approval application".

C Not here. special controls: Additional controls applicable to certain device types. The usual combination is "identify special controls".

D Correct. product code: A code used by FDA to identify a device type. The usual combination is "verify the product code".

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2. A colleague needs the term for this meaning: The date on which the reported occurrence took place. Which term fits?

B. event date

A Not here. postmarket surveillance: Systematic collection and review of information after market introduction. The usual combination is "conduct postmarket surveillance".

B Correct. event date: The date on which the reported occurrence took place. The usual combination is "confirm the event date".

C Not here. complaint: A reported concern about a device's quality, performance, safety, or related service. The usual combination is "capture a complaint".

D Not here. patient outcome: The reported effect on the patient's condition or health. The usual combination is "clarify the patient outcome".

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Review 3 / Explanations / continued

3. Rebuild the exchange from "Three participants read the label differently". Victor: "Another person finished after the moderator asked a follow-up question. The results table currently says success, with no note about the intervention." Laila: [missing response] Victor: "What about placing the status higher on the screen? That changes where attention goes, rather than only changing the text size." Which response belongs in the gap?

C. Record the prompting. We need to see how completion was achieved; assisted success should not silently become unassisted task performance.

A Not here. This line belongs to turn 6, after Victor says: "The team wants a larger font. That is easy to implement, but I cannot yet say that small characters caused all three wrong selections." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Victor says: "Three of the eight participants chose the wrong status label. The draft calls them careless. That is not something I can establish from watching the selection." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Laila's response in turn 10. It connects Victor's preceding point with Victor's follow-up. Moderator prompting can influence performance and must remain visible when interpreting task completion.

D Not here. This line belongs to turn 12, after Victor says: "What about placing the status higher on the screen? That changes where attention goes, rather than only changing the text size." It addresses that point, not the missing turn in the supplied exchange.

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Review 3 / Explanations / continued

4. Rebuild the exchange from "Incomplete information still needs a handoff". Dev: "Some records might refer to the same occurrence. We should check that before treating every entry as a separate event or comparing the clinical outcomes." Nora: [missing response] Dev: "The account is routed and you have confirmed receipt. I will correct the unknown fields and retain eight versus four complaints without inventing a rate or injury outcome." Which response belongs in the gap?

C. Yes, assess each possible duplicate report while preserving the source information. Deduplication needs a basis; a similar description alone should not make us erase a separate event.

A Not here. This line belongs to turn 6, after Dev says: "The caller did not tell us whether anyone was injured. The draft summary says no injury, which is more definite than the information actually supports." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Dev says: "The caller said the monitor stopped displaying, but the device identifier and event date are missing. I thought we should wait before sending an incomplete record onward." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Nora's response in turn 18. It connects Dev's preceding point with Dev's follow-up. A duplicate report may describe an already recorded event, so record counts are not automatically unique-event counts.

D Not here. This line belongs to turn 14, after Dev says: "The summary says device-caused. Where did that conclusion come from? The caller described the display stopping; we do not yet have the cause." It addresses that point, not the missing turn in the supplied exchange.

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Review 4 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Material identified as not meeting a specified requirement. Which term fits?

A. nonconforming material

A Correct. nonconforming material: Material identified as not meeting a specified requirement. The usual combination is "control nonconforming material".

B Not here. CAPA: Corrective and preventive action within a quality-system process. The usual combination is "assess the need for CAPA".

C Not here. supplier qualification: Evaluation and approval of a supplier for a defined purpose. The usual combination is "maintain supplier qualification".

D Not here. scrap: Material designated not to be used for its intended production purpose. The usual combination is "document a scrap decision".

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2. A colleague needs the term for this meaning: Content authorized through the relevant internal review process. Which term fits?

C. approved material

A Not here. labeling: Device-related written, printed, or graphic information accompanying or relating to the product. The usual combination is "review current labeling".

B Not here. medical information request: A request routed for an appropriate reviewed scientific or product response. The usual combination is "route a medical information request".

C Correct. approved material: Content authorized through the relevant internal review process. The usual combination is "use current approved materials".

D Not here. in-service training: Instruction delivered to professionals in their work context. The usual combination is "schedule in-service training".

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Review 4 / Explanations / continued

3. Rebuild the exchange from "Incomplete information still needs a handoff". Dev: "Is this automatically reportable because the display stopped? Service needs to avoid delay without deciding beyond its role." Nora: [missing response] Dev: "I will document the missing questions and the calls we make to obtain answers. The case should show what we tried, not simply remain blank." Which response belongs in the gap?

D. The responsible team must assess reportability under the applicable requirements. Preserve the account and route it promptly; do not replace the assessment with an automatic yes or no.

A Not here. This line belongs to turn 2, after Dev says: "The caller said the monitor stopped displaying, but the device identifier and event date are missing. I thought we should wait before sending an incomplete record onward." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Dev says: "We also need to identify the unit accurately. A model name alone might not let the investigation team distinguish this device from other units at the site." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 12, after Dev says: "I will document the missing questions and the calls we make to obtain answers. The case should show what we tried, not simply remain blank." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Nora's response in turn 10. It connects Dev's preceding point with Dev's follow-up. Reportability depends on applicable criteria and the event assessment, not the intake team's unsupported assumption.

Return to the exercise, page 100

Review 4 / Explanations / continued

4. Rebuild the exchange from "A useful question outside this session". Alex: "Then competence requires the relevant assessment; a name on the session list does not demonstrate it." Sora: [missing response] Alex: "Please route that question and tell me how we will hear back. We can continue with today's adult outpatient material while the separate response is pending." Which response belongs in the gap?

A. Correct. A competency assessment has defined criteria. We should describe exactly what was assessed and avoid extending that result to populations or tasks outside the assessment.

A Correct. This is Sora's response in turn 18. It connects Alex's preceding point with Alex's follow-up. A competency assessment evaluates defined knowledge or performance criteria beyond mere attendance.

B Not here. This line belongs to turn 10, after Alex says: "Could you still show the requested setup with a disclaimer that it is outside today's scope? I want to understand whether that would resolve your concern." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 16, after Alex says: "Thank you. While we stay with today's materials, could you clarify what the attendance certificate represents? Our team should not confuse participation with a broader qualification." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Alex says: "Could you demonstrate pediatric home use? Our team has a question about that setting, although today's examples have been outpatient adults." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 100

Additional case 1 / Explanations

Extended conversation

1. change-impact assessment

Turn 2. The change-impact assessment examines the consequences of the modification, including affected functions, risks, evidence, and regulatory questions.

2. regression testing

Turn 6. Regression testing examines whether the change adversely affects behavior that should still work, beyond the targeted defect correction.

3. configuration baseline

Turn 8. The configuration baseline identifies the controlled hardware and software versions to which the evidence applies.

4. submission assessment

Turn 10. The submission assessment determines the regulatory implications of the actual change; its outcome is not established by calling the change a bug fix.

5. deployment plan

Turn 12. The deployment plan defines the authorized update process, affected configurations, communication, and confirmation of implementation.

6. recovery procedure

Turn 16. The recovery procedure defines the assessed response to an unsuccessful update; it must not be improvised by customers or support staff.

Return to the conversation, page 103

Transfer exchange

1. version

Engineer: These results identify build 4.1, not the modified version.

Turn 1. The test evidence applies to the identified build rather than automatically covering its successor.

Additional case 1 / Explanations / continued

2. change

Manager: Assess which conclusions remain applicable after the change.

Turn 2. The modification needs an impact assessment before earlier conclusions are reused.

3. package

Engineer: Keep the open work visible in the review package.

Turn 3. The package should show the unfinished assessment instead of implying complete evidence.

4. recorded

Manager: Do not communicate release until the authorized decision is recorded.

Turn 4. A release claim requires the actual authorized decision, not an assumed dashboard status.

Return to the conversation, page 105

Additional case 2 / Explanations

Extended conversation

1. affected population

Turn 2. The affected population is the set of devices within the notice's specified scope, not simply the number of customer emails sent.

2. distribution records

Turn 4. Distribution records identify where devices were supplied and support follow-up when their current locations need confirmation.

3. Acknowledgment

Turn 6. Acknowledgment confirms receipt or recognition of the communication; it does not establish that every required device action has been completed.

4. device-level reconciliation

Turn 12. Device-level reconciliation accounts for each relevant device and its action status rather than relying on a customer-level email total.

5. effectiveness checks

Turn 16. Effectiveness checks assess whether the field communication and required response achieved their intended purpose, rather than merely counting messages.

6. closure

Turn 18. Closure is the authorized conclusion of the field action after its required evidence and conditions are satisfied.

Return to the conversation, page 107

Transfer exchange

1. completion

Service lead: The reply confirms receipt, not device-level completion.

Turn 1. Acknowledging the notice does not show that the required action was completed for each device.

Additional case 2 / Explanations / continued

2. status

Coordinator: Obtain the required device identifiers and action status.

Turn 2. Device identifiers and status are needed to reconcile the potentially affected population.

3. verified

Service lead: Keep the six units pending until the evidence is verified.

Turn 3. The reply lacks the evidence needed to establish each unit's position.

4. plan

Coordinator: Route unresolved cases through the approved follow-up plan.

Turn 4. The approved plan determines the next contact and escalation steps for unresolved cases.

Return to the conversation, page 110

Additional case 3 / Explanations

Extended conversation

1. packaging system

Turn 2. The packaging system includes the sterile barrier and relevant protective packaging; changing the outer configuration can affect protection.

2. sterile barrier

Turn 4. The sterile barrier is the packaging that maintains the sterile condition until the intended point of use, within its established conditions.

3. configuration

Turn 6. Configuration identifies the actual combination of packaging components and arrangement covered by the evidence.

4. package-integrity

Turn 8. Package-integrity evidence addresses whether the barrier remains intact; a photograph alone does not establish every relevant performance requirement.

5. shelf-life

Turn 10. Shelf-life evidence supports the claimed storage period under defined conditions; a packaging change does not automatically preserve or extend that support.

6. aseptic presentation

Turn 14. Aseptic presentation concerns making the sterile product available at use without compromising its sterile condition, not just surviving transport.

Return to the conversation, page 112

Transfer exchange

1. configuration

Engineer: The report covers the earlier packaging configuration.

Turn 1. The named insert is part of the tested configuration and differs from the proposal.

Additional case 3 / Explanations / continued

2. conclusion

Manager: Assess the effect of the new insert before reusing the conclusion.

Turn 2. The earlier conclusion cannot automatically be applied to a changed package.

3. reviewed

Engineer: Keep the change pending while the required evidence is reviewed.

Turn 3. The impact assessment and evidence review are incomplete, so approval is not established.

4. decision

Manager: Do not substitute components before the authorized change decision.

Turn 4. A proposed component replacement requires the appropriate authorized decision before implementation.

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Find a phrase by purpose

User Needs and Design Inputs

Acknowledge the need: Ease of use matters; we need to define the task behind that phrase.

Name the current audience: The agreed initial users are trained clinic staff.

Separate an expansion: Home caregivers are a proposed additional user group.

Full phrase bank, page 7

Risk Management and Hazard Analysis

Identify the source: The accessible hot surface is the potential source of harm.

Name the exposure: Prolonged contact is the exposure situation we need to examine.

Describe the outcome: A skin burn is the possible harm, not the source.

Full phrase bank, page 17

Verification, Validation, and Design Review

State the completed work: All twelve checks in this specification package passed.

Limit the conclusion: That result applies to the tested configuration and covered requirements.

Name the open evaluation: The intended-user evaluation has not started.

Full phrase bank, page 29

Usability and Human Factors

Report the observation: Three of eight participants selected the wrong status label.

Preserve the context: This occurred in the simulated task used in the formative study.

Remove blame: The notes do not establish that the participants were careless.

Full phrase bank, page 40

Find a phrase by purpose / continued

Regulatory Pathways and Submissions

State the current stage: We are evaluating a possible 510(k) pathway.

Qualify the comparator: The predicate candidate remains under review.

Correct the decision claim: No FDA decision has been issued for this device.

Full phrase bank, page 52

Complaints, MDRs, and Postmarket Signals

Capture the account: The caller reported that the display stopped unexpectedly.

Separate dates: The call was received today; the event date is still unknown.

Mark a missing identifier: We have not yet confirmed the device identifier.

Full phrase bank, page 64

Manufacturing, Suppliers, and Nonconformance

State the held population: Both lots are on hold, totaling 300 pieces.

State the observed failures: One sampled piece from each lot failed the dimensional check.

Avoid overstating scope: We have not inspected and confirmed every piece as defective.

Full phrase bank, page 80

Clinical Training and Labeling Boundaries

Acknowledge the question: That is a specific and useful question about a different population and setting.

State the session scope: Today we are covering adult outpatient use with version 4 materials.

Name the boundary: These materials do not cover pediatric home use.

Full phrase bank, page 91

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References and publication notes

All cases, organizations, figures, and conversations are fictional teaching material, not transcripts. Communication practice does not replace qualified advice or current workplace procedures.

[1] US FDA. Quality Management System Regulation (QMSR).

Current US framework, effective 2 February 2026, incorporating ISO 13485:2016. Legacy quality-system terminology should not be mistaken for the current rule. Source checked 10 October 2026.

https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr

[2] US FDA. Applying Human Factors and Usability Engineering to Medical Devices.

Background on intended users, environments, interfaces, and use-related risk. All study observations and proposed changes in this book are invented. Source checked 10 October 2026.

https://www.fda.gov/regulatory-information/search-fda-guidance-documents/applying-human-factors-and-usability-engineering-medical-devices

[3] US FDA. Medical Device Safety and the 510(k) Clearance Process.

Background on pathway distinctions and the comparative 510(k) review standard. Fictional cases do not determine a device classification or submission route. Source checked 10 October 2026.

https://www.fda.gov/medical-devices/510k-clearances/medical-device-safety-and-510k-clearance-process

[4] Electronic Code of Federal Regulations. 21 CFR 803.50.

US manufacturer reporting provisions, including reasonably known and missing information. Apply all relevant provisions and specialist review, not a classroom deadline. Source checked 10 October 2026.

https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-803/subpart-E/section-803.50

> **Companion course**
>
> The website offers the eight-lesson course and related practice. This book can be used independently. No audio access or external account is needed for its exercises.
>
> https://englishladder.com/efsp-medical-devices.html

Specify the need. Explain the evidence. Keep the boundary clear.

Eight conversations across device development, postmarket information, and responsible product communication.

User Needs and Design Inputs

A proposed two-minute target is not an approved acceptance criterion. The team must identify what setup includes, who performs it, under which conditions, and why the target is appropriate before treating the number as a settled requirement.

Risk Management and Hazard Analysis

Hazard, hazardous situation, and harm are not interchangeable. In this case, the potentially hot surface is the source; prolonged contact describes exposure; a burn is the possible injury. These distinctions clarify the analysis without supplying a likelihood estimate.

Verification, Validation, and Design Review

The twelve checks support the specified requirements they address, in the tested configuration. They do not automatically cover every user need, later version, or intended-use condition. Report completion at the level supported by the actual evidence.

Usability and Human Factors

Three of eight is 37.5% of this study group, not an established error rate for the user population. Report the count and study context. A formative finding can identify a design concern without providing a final validation conclusion.

Phil Stilwell

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